A Realistic, Normative, General (Janus) Equilibrium for Asset Prices and Wages

Preprint · December 2024

DOI: 10.13140/RG.2.2.27594.89282

1 author:

Arun Muralidhar
Georgetown University

A Realistic, Normative, General (Janus) Equilibrium Model for Asset Prices and Wages

By

Arun Muralidhar©

Original Version: December 2024; this version August 2025

Abstract

A realistic, normative, general equilibrium model, where individuals achieve multiple, stochastic investment goals, but with "implicit" utility, production and work allocation functions, because we use actual objectives of investors. Additionally, the successful creation of innovative retirement and education bonds by Brazil, shows that the existence of just two of these securities, plus the absolute risk-free asset, determines the price of all assets and provide effective asset allocation recommendations, with no free parameters. We demonstrate reasonable prices for securities, given live correlation and volatility data, and, in turn, the wages that must be earned by individuals to hold such portfolios.

Keywords and JEL Classification

Keywords: Arrow-Debreu Securities; Asset Pricing; Agency; Luck versus Skill; Goals and Risk-based Asset Pricing Model; Relative Asset Pricing Model; Stochastic Goals; Liability Driven Investing; Dual Normalization; Chameleons; State Prices; (No) Free Parameters; Goals-based Arrow-Debreu Securities; Heterogenous Investors; Janus Equilibrium; View Neutral Allocation; Goals-Based Investing; General Equilibrium; Utility Maximization; Implicit Production Function, Implicit Labor Allocation

JEL Classification: G1, G2, G12, C30, D21, D5, D14, J22, J3

Disclosure Statement

I have no conflicts or funding issues to disclose.

1. I received no financial support from any entities
2. There was no third-party review required or conducted of this paper
3. There were no non-disclosure arrangements that have any relevance to this paper.

"Moreover, since the sun remains stationary, whatever appears as a motion of the sun is really due rather to the motion of the earth."

"*Those things which I am saying now may be obscure, yet they will be made clearer in their proper place*." *Nicolaus Copernicus*[1]

Introduction

Arrow (1953), Debreu (1959) and Arrow-Debreu (1954) provide a general equilibrium (GE) model that uses the concept of "state prices", also known as Arrow-Debreu (AD) securities, to prove the existence of a unique general equilibrium (GE). They assume a very abstract model of utility maximizing individuals, profit maximizing firms, and a fixed number of "commodities" to establish this competitive equilibrium. Despite its popularity in every finance program, the model fits the description of Pfleiderer's (2020) "chameleon" given the unrealistic assumptions to derive this GE. The AD model is commonplace in the asset pricing literature as the idea of a security that pays 1 unit in a particular "state" and zero in all others, is a neat construct and allows for the pricing of all assets (Arrow 1953 even refers to these as "elementary" securities). But this AD asset pricing model is insufficient, because investors require a consistent asset pricing model, asset allocation recommendations and risk-adjusted performance measures (or the "three facets of investing") to be effective in managing portfolios as discussed in Muralidhar (2019, 2023); the Arrow-Debreu model is totally silent about asset allocation and risk-adjusted performance. Moreover, they do not specify a realistic objective function of the consumer/investor and hence the model is much too abstract to be practical.

We demonstrate instead, that if we start with four realistic positive observations about the investment environment, we can present a normative GE model where ALL assets can be priced with no "free parameters" (Cochrane 2005) with just: (a) two "goals-based Arrow-Debreu securities" (GADs), as described in Muralidhar (2024a) and below; and (b) the traditional risk-free asset. Muralidhar (2021, 2023, 2024) describes this asset pricing model and labels it the Goals and Risk-based Asset Pricing Model (GRAPM); the "no free

[1] https://www.brainyquote.com/authors/nicolaus-copernicus-quotes

parameter" solution follows because of the duality of many equilibrium equations in GRAPM, leading it to be termed a "Janus Equilibrium". Interestingly, adding more GADs provides more equilibrium equations and a more compact GE.[2] However, since we also know the asset allocation of investors/consumers/workers (which AD ignore), we also know the wage that they must earn to satisfy the budget constraint of purchasing the GADs and other assets for their respective goals (i.e. the value of the portfolio they must buy to meet their objective function). This then allows us to assume an "implicit" production function of firms/entities that produce these goal-related cash flows, and pay wages (and earn a zero economic profit), such that the value of the firms/entities[3] is represented by the GADs prices. We can also impute a work allocation function that these investors/workers must apply to split their time working among the various entities that issue these securities.

GRAPM is based on four realities of investing: (i) that investors have stochastic goals (and not a deterministic goal as assumed in Sharpe 1964); (ii) that investors have multiple stochastic goals (retirement, education, health etc. unlike single goal models in the literature); (iii) that they delegate to (hopefully) skillful agents; and (iv) they specify their objectives, especially their risk preferences, in a very focused/transparent manner, with clear absolute and relative risk goals. GRAPM utilizes the M-cube risk-adjusted performance measure in Muralidhar (2000), and not a traditional utility function (which could also be imputed from the risk-adjusted performance measure). The objective function in GRAPM is taken from the Investment Policy Statements (IPSs) of real investors, incorporating delegation to agents, and a desire for skillful agents. Hence, we treat the utility function as "implicit" and we need not assume an arbitrary utility function as is the case in the finance literature. This approach is in line with how Perold (2004) establishes the duality of the Capital Asset Pricing Model (CAPM), by demonstrating it can be solved by either the utility function approach of Sharpe

[2] Haugh and Lo (2001) show that derivative securities are equivalent to specific dynamic trading strategies in complete markets and hence suggest the possibility of constructing buy-and-hold portfolios of options that mimic certain dynamic investment policies, e.g. asset-allocation rules. In a way, they integrate asset pricing and asset allocation from a dynamic trading perspective. I thank Jean-Luc Vila for this point.
[3] We use a generic term entity as certain securities, issued by governments, like the absolute risk-free asset, could also be priced. In other words, issuers of securities in this model need not just be profit maximizing firms, but also welfare maximizing governments.

(1964) or the maximizing risk-adjusted performance of Sharpe (1994). The nice thing about this approach is that it also lends itself to a range of general equilibrium model applications, including a Preferred Habitat Theory of Asset Pricing/Term Structure (Muralidhar 2025a), an Entangled Markets Hypothesis (Muralidhar 2025b), and a realistic, Heterogeneous (Principal-) Agent Model of Asset Pricing (Muralidhar 2025c).

The GE model presented is further grounded in reality as Brazil issued two such GADs in 2023 – RendA+ (or "retirement income bond")[4] and Educa+ ("education bond")[5], based on Muralidhar, Ohashi and Shin (2016) and Muralidhar (2016), respectively. These bonds correspond to the Cochrane (2022) observation that, "Switching to a payoff perspective may change lots of procedures in finance". As will be shown later, a critical issue for the GE model is that for retirement investors, Educa+ is risky (as is the absolute risk-free asset of the CAPM - Merton 2014), and for education investors, RendA+ (and the absolute risk-free asset) is risky. Additionally, RendA+ tailored to a 60-year-old is risky relative to RendA+ tailored to a 25-year-old (very easily understood by the reader given the duration mismatch of the cash flows). This is the "payoff perspective" at work, because each stochastic goal payoff is unique.

Going back to the AD elementary securities, what these GADs offer in their design, as articulated in Muralidhar (2016), is the fact that rather than paying 1 unit in a particular state, they pay a unit of a series of cash flows tailored to be the exact cash flows needed for the goal. This makes sense as life-cycle models in economics, as initially shown in Modigliani and Ando (1963), and much more forcefully in Merton (2014), is all about managing cash flows, and not wealth. In fact, one could see the desire for such sustained cash flows, on a forward start date, as a form of "Habit Formation" or Keeping Up with the Joneses" utility functions, so common in the asset pricing literature. But this is more focused on the reality of investing.

[4] https://borainvestir.b3.com.br/tipos-de-investimentos/renda-fixa/tesouro-direto/tesouro-renda-brasil-pode-ter-um-sistema-previdenciario-mais-sofisticado-que-o-de-outros-paises/
[5] https://www.cnnbrasil.com.br/economia/tesouro-educa-novo-titulo-para-financiar-estudos-comeca-a-ser-negociado-na-b3-veja-como-funciona/.

As a result, buying one unit of RendA+ or Educa+ is equivalent to an AD-type perspective leading to them being called GADs. For retirement, a GAD would offer \$10 real/year for say 20 years (average life expectancy), starting at the retirement age, and ideally indexed to standard-of-living (Bodie, Merton and Samuelson 1992). Education GADs (also called Bonds for Education and Student Tuition or BEST) would offer \$10 real/year for say 4 years (5 in the case of Educa+), starting at age 18 (when a child goes to college), and indexed to tuition inflation. The cash flows of generic Brazilian retirement and education bonds are provided in Figure 1, but there is no requirement that these GADs be solely debt instruments, as certain goals have equity-like components.

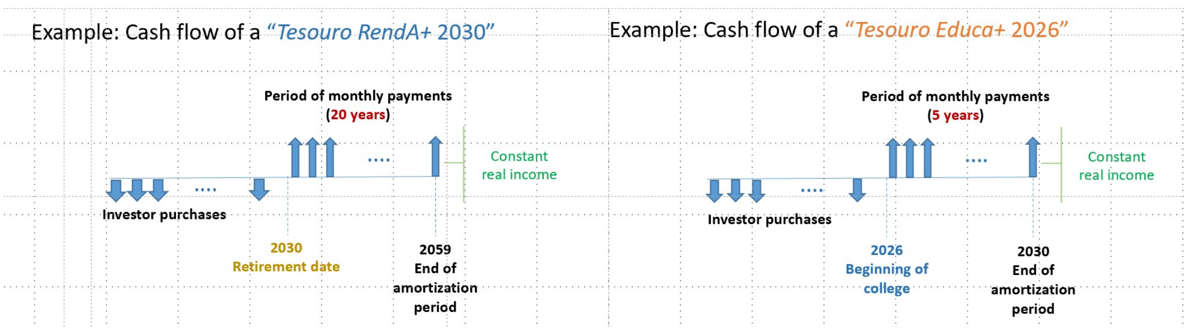


Figure 1: Cash flows of retirement (20 years starting in 2030), and education (5 years starting in 2026) bonds issued in Brazil (*Source: Tesouro National presentation to International Retail Debt Management Symposium, Washington DC, November 2024*).

If an individual needed \$50,000 real/year in retirement or for a child's college education, they just need to buy 5,000 of the respective bonds prior to the trigger event (with the flexibility to change their date as these are liquid and tradeable on a simple app offered by Brazilian Treasury). In short, to achieve one's goal requires buying just multiples of a single, relative risk-free security (Merton and Muralidhar 2020), GADs, as opposed to current practice of buying multiple securities to match goal cash flows (Coqueret et al 2017). This is how GADs can greatly improve goal-achievement and risk management in a goals-based world. But, the incentive for the issuer to create this instrument implies either an implicit production function (in the case of a firm) or some welfare function (as the issuer could be the government that seeks a social goal), such that the application of capital, and paying appropriate wages, offers the established rate of return derived in the asset market equilibrium. Hence, this satisfaction of desired (multiple) individual cash flows like retirement and education by entities for whom it is profitable to provide these cash flows (whether for profit maximization or general social welfare), gives us a normative GE model, grounded in positive observations of investors and issuers, and gives us equilibrium asset prices and wages, that do not require forced utility, production or labor allocation functions.

This paper is different from GE models of say Merton (1973) that models an Intertemporal Capital Asset Pricing Model (ICAPM), but ignores the real economy and treats the investment opportunity set as stochastic. In reality, the goal for investing is the primary stochastic variable that drives investment decisions, as the burgeoning goals-based investing (GBI) literature has shown, and not the stochasticity of the opportunity set. Interestingly, Merton (1993) indirectly acknowledges the stochasticity of a goal in driving investment decisions for a university endowment and more recently, Merton (2014) does so in the context of defined contribution investing, arguing that it is not wealth that matters in DC plans, but retirement income! Lucas (1978) starts with an "endowment economy", which includes an arbitrary production process and a perishable good, to derive asset prices, but this is an abstract model, much like the AD GE model. Cox, Ingersoll and Ross (1985), CIR hereafter, who similarly model a real and financial economy, but the model is based on forecasts of real variables, which are notoriously difficult to identify or forecast. More recently, Gomes and Schmid (2017) offer a GE model "linking the pricing of stocks and corporate bonds to endogenous movements in corporate leverage and aggregate volatility." Other GE models like Ai, Croce and Li (2013), with "investment options as intangible capital in a production economy", do not have similarities to our approach which starts with positive observations about reality to derive normative models of asset pricing and asset allocation.

The paper is structured as follows because we believe we achieve Cochrane (2022), when he states, "portfolio theory and application should describe a general equilibrium, and how investors are heterogeneous." Section I highlights key observations from Arrow-Debreu (1954) to motivate this discussion (including the AD critique of previous GE models, but also to address some key issues noted by Cochrane 2022), and expands on the four key realities that provide the basis for this model. Section II then lays out the historical background on financial innovation and the two key securities created by Brazil as that is foundational to our model because we show that investing is all about the cash flows, and heterogeneity is established by a desire for different cash flows for different goals. We demonstrate how focusing on cash flows desired by investors (demand-driven) is different from previous research that focused on the cash flows suited to the issuer (supply-driven), and that have largely not succeeded. Section III sketches out the real and financial economy we will model to show that it is basically a "financing model"; individuals wanting to finance consumption needs; firms needing to finance an implicit production function. Section IV provides the specific assumptions needed to establish the GE result based on individual's meeting their consumption/investment goals, subject to the budget constraint, and entities (to include firms and governments) to achieve profit/welfare maximization. Section V first establishes the individual optimization equilibrium prices, but also the wages that must be earned, on the basis of the budget constraint, to ensure investors achieve their investment objectives. We use live data from 8/1/2023-10/29/2024 of Educa2030, RendA2060 and the S&P500 Total Return Index (in Brazilian Real terms), to show what a normative model would require, relative to live data, and to provide an equilibrium set of expected returns and wages, going forward. Adding more GADs changes the equilibrium prices obviously but this is left to Appendix VII. Section VI introduces the entity perspective to demonstrate what results an "implied production function" must satisfy for profit maximization, given equilibrium prices of assets sold and wages earned in aggregate by the investors (and hints at the work allocation decision of individuals to work among these entities to earn sufficient wages to buy their optimal portfolio). Section VII compares the AD approach to GADs for the GE to demonstrate its applicability and suggests extensions. Section VIII concludes.

I. The AD Model, and Four Key Realities of Investing

In this section we provide additional detail about the AD model and the reason they felt their model dominated others at that time to compare our own model later in Section VII. We also provide more color on the four specific realities of investment that drive our model as that provides context to the academic reader, not familiar with actual practice of institutional investors (that we use as the "representative" investor because their practices are easier to monitor and more uniform across the globe).

A. *The Arrow-Debreu Model Assumptions and Conclusions*

AD (1954) argue that, "It is well known that, under suitable assumptions on the preferences of consumers and the production possibilities of producers, the allocation of resources in a competitive equilibrium is optimal in the sense of Pareto (no redistribution of goods or productive resources can improve the position of one individual without making at least one other individual worse off), and conversely every Pareto-optimal allocation of resources can be realized by a competitive equilibrium…. Here proofs of the existence of an equilibrium are given for an *integrated* model of production, exchange and consumption…. A central concept is that of an abstract economy, a generalization of the concept of a game."

Their abstract model is based on, "a finite number of distinct commodities (including all kinds of services). Each commodity may be bought or sold for delivery at one of a finite number of distinct locations and one of a finite number of future time points…. The commodities, or at least some of them, are produced in *production units* (e.g., firms)…..Under the usual assumptions of perfect competition, the *economic* motivation for production is the maximization of profits taking prices as given…. $p*$ denotes the equilibrium price vector. The above condition is the first of a series which, taken together, define the notion of *competitive equilibrium.* Analogously to production, we assume the existence of a number of *consumption units,* typically families or individuals but including also institutional consumers….. to have equilibrium, it is necessary that each individual possess some asset or be capable of supplying some labor service which commands a positive price at equilibrium…The basic economic motivation in the choice of a consumption vector is that of maximizing utility among all consumption vectors which satisfy the budget restraint, i.e., whose cost at market prices does not exceed the individual's income."

They demonstrate superiority over previous models by arguing that, "Cassel (1924)….four basic principles of his system: (1) demand for each final good is a function of the prices of all final goods; (2) zero profits for all producers; (3) fixed technical coefficients relating use of primary resources to output of final commodities; and (4) equality of supply and demand on each market. Wald (1951) assumes fixed proportions among the inputs and the single output of every process. On the demand side, he makes assumptions concerning the demand functions instead of deriving them, as we do, from a utility maximization assumption." We will return to these key points in Section VII by comparing our model results with the AD model (and even tangentially, the Lucas and CIR model), especially since the inputs in our model are based on the actual performance of assets/GADs currently traded in the markets.

B. Four Realities of Investing

The AD model was developed well before an active financial industry, especially institutional investing, existed. The four key realities of investing that form the anchor of our model, and taken from the practice of institutional investing, was probably non-existent in the 1950s, as opposed to the abstract AD model. While we use American examples below, these practices are global and reasonably similar, with each country having their own nuance because of regulatory and institutional issues. The four observations we make that drive our four realities of investing in the Introduction are:

- Many Stochastic Goals: Individuals (and even institutions) invest to achieve many stochastic goals. By stochastic goal, we mean that the present value of the stream of cash flows required for the goal can change daily because of changes in market parameters, including interest rates, and various types of inflation (standard-of-living, tuition or health, respectively), or the occurrence of an event. Stochastic goals range from retirement (Merton 2007), to saving for a child's college education (Muralidhar 2016), to saving for future health expenses, among many others. Since they are custodied separately for regulatory reasons, one cannot run a global optimization for asset allocation (as in Das, Markowitz, Scheid and Statman 2010), but moreover, the diversity of goal cash flows leads to interesting challenges. Similarly, some institutional investors manage multiple portfolios under the same governance structure and team: endowments, one or more pension funds depending on the beneficiaries (academic versus support staff), retiree health benefits, and even insurance portfolios as in the case of say the University of California. Sharpe (1964) by contrast, implicitly assumes that his single goal is deterministic (and discounted by the absolute risk-free asset, *F*)[6]. Interestingly, Sharpe-Tint (1990), whose objective statement we use in GRAPM, acknowledges the importance of (a single) stochastic goal in setting asset allocation, but never used the same model to derive asset pricing (as was shown in Muralidhar, Ohashi and Shin 2014a and 2014b, in a model they call the "Relative Asset Pricing Model" or RAPM, using a KUJ utility function with the stochastic liability as the denominator and the asset portfolio or wealth in the numerator).

- Investor Objectives: Investors seek to maximize goal-relative, net, expected risk-adjusted returns, and not maximize utility of wealth (or maximize expected returns) as assumed in a lot of the literature, especially the CAPM and ICAPM models, but also AD's GE model.[7] The Los Angeles County Employees' Retirement Association's (LACERA) IPS explicitly states: "The Fund's long-term performance objective is to generate risk-adjusted returns that meet or exceed its defined actuarial target as well as its policy benchmark, net of fees, over the Fund's designated investment time horizon."[8] The actuarial target proxies the stochastic goal of this pension fund (i.e., the pension benefit payments). This is not an isolated statement of objectives, but found in probably every pension fund and every well managed endowment portfolio IPS. After all, wealth for wealth's sake has no meaning (Merton 2014), and even Socrates is believed to have stated that, "If a man is proud of his wealth, he should not be praised until it is known how he employs it".[9] Cochrane (2022) hints at this by stating, "We start the statement of any investment problem as a search for the optimal stream of payoffs," but most investors first convert payoffs into a proxy portfolio whose returns and risks can be tracked daily (Muralidhar and van Stuijvenberg 2005).

---

[6] Prof. Sharpe has acknowledged this in private correspondence with the author.
[7] See for example, Modigliani and Modigliani (1997), Muralidhar (2000), Perold (2004) and Muralidhar (2023).
[8] https://www.lacera.com/sites/default/files/assets/documents/general/invest_policy_stmt_031921.pdf - page 8.
[9] https://www.azquotes.com/quote/669505.

So, Cochrane (2022) has the order reversed. Required payoffs for goals determine asset allocation and in turn prices of assets (and wages to finance portfolio purchases).

- Delegation: Investors delegate investment decisions to agents (Allen 2001), but seek skillful agents (Ambarish and Seigel 1996). The literature has focused more on the risk budget and benchmarks allocated to agents (van Binsbergen et al 2007), rather than ensuring agents are skillful in asset pricing models. Even when focused on asset pricing, Brennan (1993), or Cornell and Roll (2005) among many others, writing on agency-based asset pricing, ignore the importance of skillful agents and assume the agents can take any degree of relative risk they want, whereas as we will show below, the IPS (or Board reporting) is very specific about the degree of relative risk permitted.
- Explicit Statement of Absolute and Relative Risk Preferences: Investors explicitly specify their risk budget in their IPS in terms of absolute and relative risk targets. For example, New Mexico Public Employees' Retirement Association's (NMPERA) IPS explicitly states that the Board established a 10.5% annualized target volatility for the strategic asset allocation (absolute risk) and a 1.5% annualized tracking error (or TE hereafter) for all delegated decisions (as tracking error measures relative risk).[10] Similarly, LACERA, like NMPERA, also articulates an explicit relative risk budget. Even without explicit risk budgets (as in the case of say the Maryland State Retirement Agency or SRA), staff report an implicit risk budget in their periodic reports to their Boards.[11] So, risk aversion is explicitly stated in these parameters and transparent reporting, and not an arbitrary, unobservable parameter in a non-observed utility function.

Interestingly, it is not just the academic literature that misses these nuances. Even the ubiquitous Black-Litterman model (Black and Litterman 1991, 1992) makes no mention of a single stochastic liability, let alone multiple goals/liabilities, and ignores agency issues to derive a "global equilibrium" to help internal asset managers and pension

[10] See for example the statement of objectives of the New Mexico Public Employees' Retirement Association. http://www.nmpera.org/assets/uploads/downloads/RIO/RFP/RFP-NO.-NM-INV-001-FY19-Total-Fund-Overlay-Services.pdf.
[11] https://sra.maryland.gov/sites/main/files/file-attachments/2022q4_performance_report_redacted.pdf?1680025511. See page 22.

funds/endowments/sovereign wealth funds make short-term or long-term asset allocation decisions. Ignoring the realities and the relativities of markets in making such decisions cannot bode well for say retirement security and any "outperformance" is probably more due to luck than a true representation/skillful market-outperformance view because of effective asset allocation. Cochrane (2022) believes institutional investors do not ask academics for advice for fear of being told they are wrong; quite the contrary, the models adopted by academics are Pfleiderer "chameleons", and hence divorced from reality and practical use by investors. This paper seeks to address this challenge to start from practice to derive theory that can then improve practice.

Given the volatility and TE targets, it becomes critical to ensure that dual normalizations of volatility and correlation take place for effective asset pricing, allocation and risk-adjusted performance calculations, otherwise principals could be making sub-optimal decisions based on poor approaches; namely, hiring lucky agents. This will be demonstrated in Section IV.

II. A Sidebar on Financial Innovation for Goals-based Investing

The challenge with the AD securities and the AD GE model is that "states" are not defined, and there is no obvious issuer for each AD security as the "states" are abstract and not specified in the original model. While the original papers talk about futures prices, for specific quantities, delivered on a particular date and a particular location, the classic AD security paying 1 unit in a particular state and zero in all others (basically a Dirac delta function) is impractical. But then, the AD model is based on assuming that investors maximize a utility function, of a very particular form (preferences that are continuous, non-satiated, and convex), and with an unobserved "risk aversion" parameter. On the other hand, Muralidhar (2016) changes the paradigm to focus on satisfying the cash flows needed for a range of goals (as proposed in Merton 2007) as opposed to wealth, as that is what is what is truly relevant for every investment portfolio. In a GBI approach, one must start with the stochastic cash flows associated with the goal (as a way to benchmark that goal on a daily basis) as in Figures 1 or 3. In a GBI world, cash flows dictate the investment decisions.

There has been a history of (academic) recommendations for creating new financial instruments, not always ending up with a successful launch, in part because they were based on academic theory or because they were designed from the perspective of the issuer. Merton (1983) had argued for the creation of a unique pension bond that should be issued by the government, indexed to standard-of-living (e.g., per capita consumption), and sold to the Social Security Administration, so that social security pensions could be indexed to standard-of-living and not just inflation. This innovative idea has been modified in the design of SeLFIES (Standard-of-Living, Forward-starting, Income-only Securities of Muralidhar and Merton 2020, and in part the model on which Brazil's RendA+ was launched). The indexation of Merton (1983) was critical, not the relatively unspecified cash flow, that was recast and more clearly specified in Muralidhar, Ohashi and Shin (2016). Shiller (1993, 2005) advocated for the creation of GDP-linked bonds, which while useful for the issuer (supplier), because it allowed for lower debt service during periods of economic stress, had no obvious demand for the cash flows from institutional or retail investors (who are already long GDP growth in their equity allocations or their individual lives), and hence were largely unsuccessful despite being issued by a few European countries. Roll (1996) discusses the design of Treasury Inflation Protected Securities (TIPS), with a cash flow driven more by preserving real wealth than directly linked to a stochastic goal (which is feasible to achieve with a series of zero TIPS STRIPS, with extremely complex financial engineering, but this dramatically increases the cost, illiquidity and risks of using such instruments). The US version of these instruments have not been as successful as hoped as private pensions in America are not inflation-indexed; the UK inflation-linked gilts ("linkers") have been more successful.[12]

Following Merton (2014), Muralidhar (2016) highlights that for most (retail) goals and some institutional goals, these goal cash-flow replicating securities do not exist, and hence argues for the creation of such instruments in both debt and equity markets. In effect, by focusing first on the demand side of such cash flows (as opposed to the supplier side as in Shiller (1993)), there is a slightly higher likelihood of success. In order to create such instruments,

[12] Despite having been launched in 1997, these securities have not been as big a success leading Treasury official to recently enquire of the author of this paper as to why TIPS have failed. https://www.treasurydirect.gov/research-center/history-of-marketable-securities/tips/

one must find a natural issuer for these instruments so that supply meets demand. Muralidhar (2016) shows how a natural issuer for the retirement goal would be a government, and a natural issuer for education could be private institutions (especially in the US, dominated by private colleges and universities in need of funds above and beyond tuition receipts)[13] or governments, and so on for other goals. For example, the natural issuer for the absolute risk-free asset would also be governments.

Brazil issuing the RendA+ retirement bond and Educa+ education bond, with significant and growing demand (over 100,000 investors voluntarily buying each of the securities in less than 2 years), not only proves this point, but the fact that other countries are considering these innovations[14] shows that these are global issues and not just isolated to Brazil. Muralidhar (2018) extends this idea of GADs further to show how cohort-level longevity risk can be hedged in retirement accounts with a natural issuer (governments). Other research has highlighted how corporate entities could issue financial securities to hedge other goal cash-flow risk. The existence of entities to issue such debt or equity securities addresses the critical assumption in this paper that GADs not only have demand from investors, but have a credible supplier with an offsetting hedge.

III. A High-Level Production and Consumption Economy

Figure 2 captures the economy that we are modeling with two major actors. The primary actor is "individuals", who work for "entities" (the second) to earn a wage so that they can they achieve their stochastic investment/consumption goals. For simplicity, we assume that there are only two goals (say retirement and child's education), and two individuals/workers/consumers (and they can delegate the investments to each other for no fee for simplicity and it doesn't prevent them from working for the entities). We could relax this assumption later without loss of generality but it keeps the model simple. The

[13] Estimated to be above $230 bn in total - https://www.highereddive.com/news/merger-watch-bondholders-college-closures/716524/#:~:text=In%20fact%2C%20a%20recent%20Financial,billion%20of%20outstanding%20bond%20debt.

[14] See https://selfiesforall.wordpress.com/2019/02/05/country-op-eds/

“commodity” that they consume is the unique set of cash flows (CF) needed to achieve their goals. These specific CFs are produced by two entities (say E_{G1} and E_{G2}), but there are also two more entities: one, say a government, that issues *F* (the absolute risk-free asset) and potentially with a welfare maximizing rather than profit motive; the other produces a risky cash flow, *I*, that is risky for both goals. So, this model can be seen as a GE financing model with entities needing to finance their “production” activities (with an unspecified production function), and individuals needing to finance their desired consumption (with a clear objective function, but unspecified utility and labor allocation functions). Despite this lack of specification of the various functions central to traditional economic and finance theory, we get very specific prices for securities and conditions for equilibrium wages.

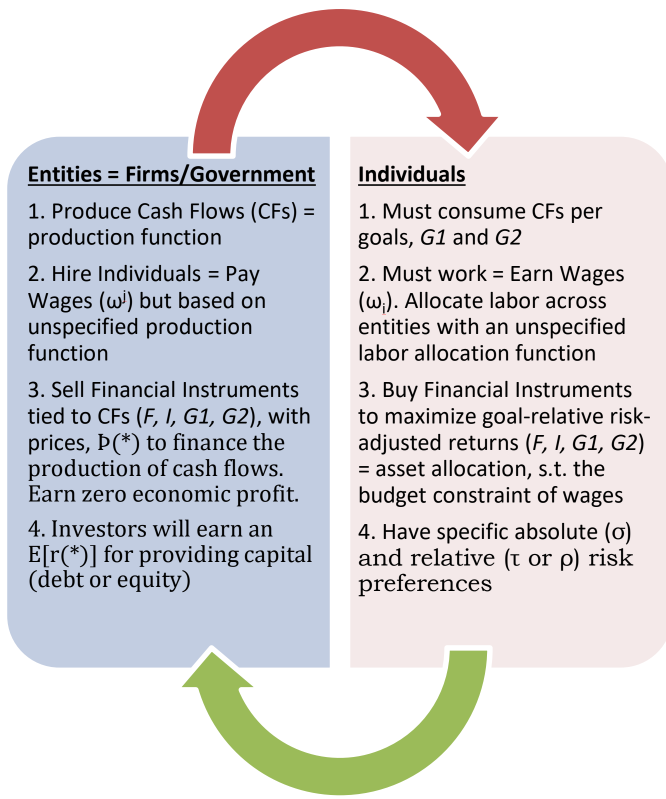


Figure 2: The Circular Economy of Production and Consumption of Cashflows

The nomenclature followed and high-level (stylized) economy design will be

1. **Production Function Set Up – Entities Generating Cash Flows and Hiring Workers (Implicit Production Function).**
   a. Assume that there are entities (*E*) that produce goods and services/ "commodities". We do not care about these commodities particularly, but rather the CFs that these entities generate from the sale of their commodities/services.
      i. Assume we have four such entities (and the government could be multiple entities as in the case of Brazil)
      ii. E_{G1} – an entity that generates cash flows = *G1* (e.g., an infrastructure firm with large capital outflows today for a stream of income for 20 years, starting 35 years from now in 2060).
      iii. E_{G2} – an entity that generates cash flows = *G2* (e.g., a university with large capital outflows today for a stream of income for 4-5 years, starting 5 years from now in 2030 – i.e., for one's 13-year-old child)
      iv. E_F – an entity that has short term funding needs = *F* (e.g., a government that needs to financing to fund itself with say T-bills). *F* is the absolute risk-free asset of CAPM and ICAPM.
      v. E_I – an entity that generates stochastic cash flows = *I* (e.g., a single company or an entire stock market as in CAPM and ICAPM).
   b. Goal Cash Flows (*G1* and *G2*) can be visualized in Figures 1 or more specifically, Figures 3A and 3B below, and note this is identical to the production function cash flows (i.e., cash flows created by the production technology).
   c. From Modigliani-Merton (1958), we can assume that the value of the firm is the discounted present value of real cash flows, and not how the firm is financed, so we ignore firm financing structure. Hence, the claims on cash flows are claims on the firm.
   d. If we assume that *E[r(*)]* is the expected return of asset *r(*)*, in this simple one-period model, the price, Þ, of these securities will be given by
      i. $\text{Þ}(\boldsymbol{G1}) = \frac{\mathbf{1}}{(\mathbf{1}+\boldsymbol{E}[\boldsymbol{r}(\boldsymbol{G1})])}$

ii. $\text{Þ}(\boldsymbol{G2}) = \frac{\mathbf{1}}{(\mathbf{1}+\boldsymbol{E}[\boldsymbol{r}(\boldsymbol{G2})])}$

iii. $\text{Þ}(\boldsymbol{F}) = \frac{\mathbf{1}}{[\mathbf{1}+\boldsymbol{r}(\boldsymbol{F})]}$

iv. $\text{Þ}(\boldsymbol{I}) = \frac{\mathbf{1}}{(\mathbf{1}+\boldsymbol{E}[\boldsymbol{r}(\boldsymbol{I})])}$

e. In effect, to establish the value of the entity (i.e., the assets they sell against their respective cashflows), we will solve for *E[r]*

f. There are two workers – Sid and Sachin – who work for these entities to earn wages to satisfy the goals-based cash flows implicit in their utility/consumption function.

i. Sid has the *G1* goal = S_{G1}

ii. Sachin has the *G2* goal = S_{G2}

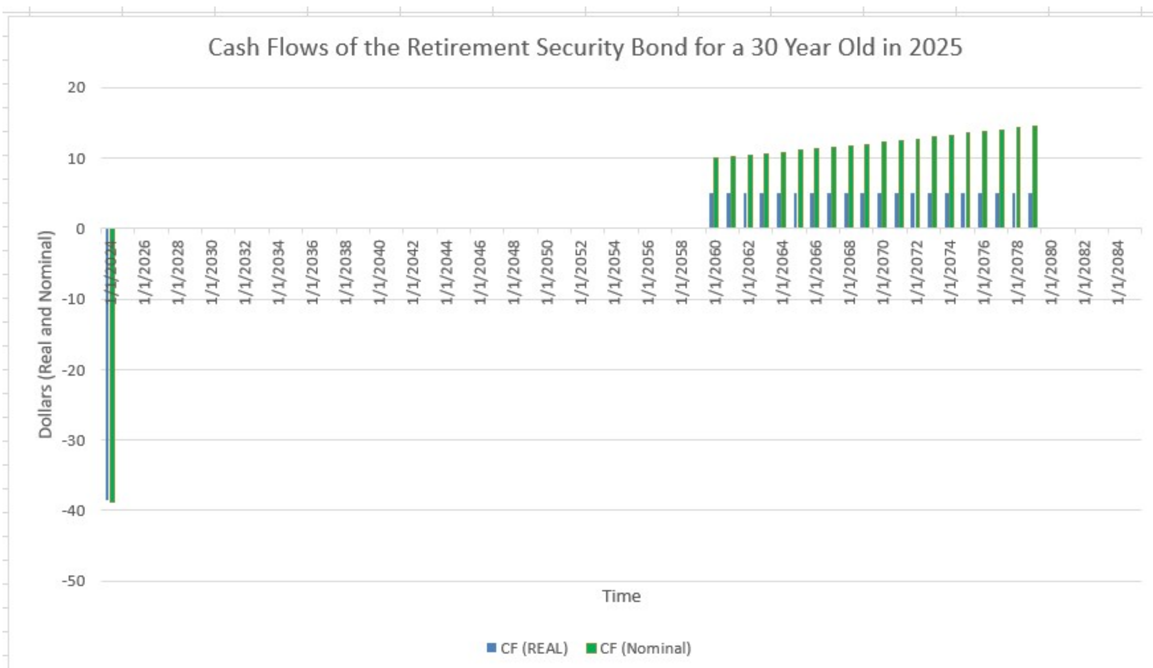


Figure 3A – Real and Nominal G1 Cash flows (assuming a 2% p.a. inflation). Initial negative bar = price today (Dec 2024). Call this RendA2060

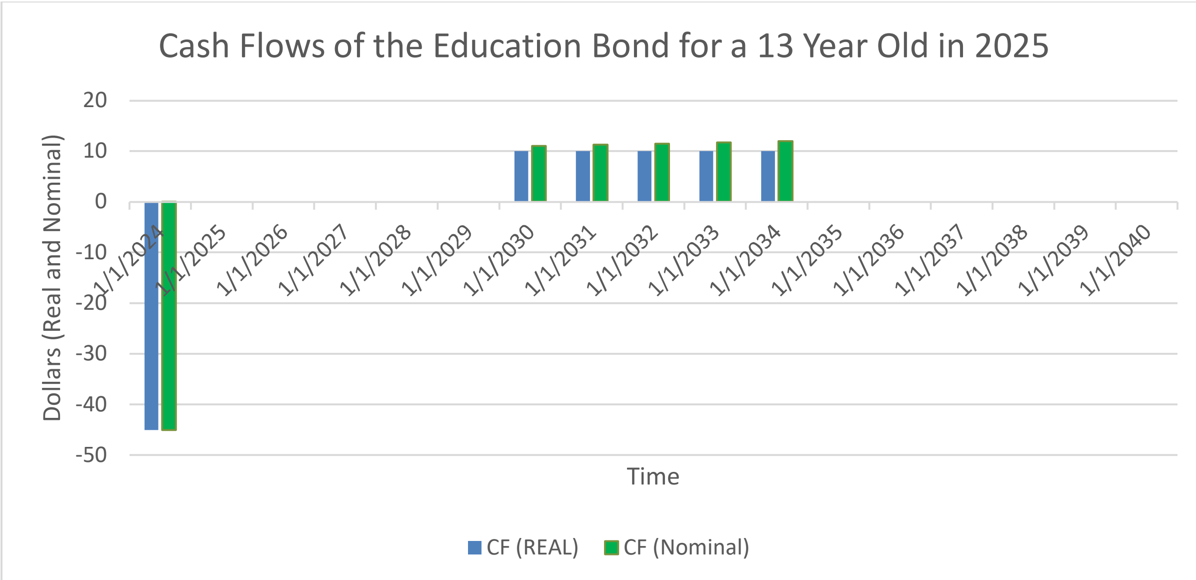


Figure 3B – Real and Nominal G2 Cash flows for 5 years (assuming a 2% p.a. inflation). Initial negative bar = price today (Dec 2024). Call this Educa2030.

g. Wage Payments:

i. E_{G1} pays wages ω_{G1}^{G1} to *G1* investor (Sid) and ω_{G2}^{G1} to *G2* (Sachin), where the lowercase refers to investor/worker (earning wages) and the upper indicates the entity (paying wages);

ii. E_{G2} pays wages ω_{G1}^{G2} to *G1* investor and ω_{G2}^{G2} to *G2*; and

iii. E_I pays ω_{G1}^{I} and ω_{G2}^{I},

iv. Technically, we could do the same with E_F, but for simplicity, we will assume that the government entity is a passthrough and does not pay wages (see below).

h. Wages Earned: For simplicity, we will ignore the labor-leisure choice function, or allocation of work between entities, but could easily add that into the model in the future, with some arbitrary labor allocation model that fits the GE model. Notice again, we will ignore wages from E_F,

i. S_{G1} earns $\omega_{G1} = \omega_{G1}^{G1} + \omega_{G1}^{G2} + \omega_{G1}^{I}$ and

ii. S_{G2} earns $\omega_{G2} = \omega_{G2}^{G1} + \omega_{G2}^{G2} + \omega_{G2}^{I}$

i. Implicit Production Function: Entity E_{G1} has a production function = $\emptyset_{\boldsymbol{G1}}(G1, ꓘ, \omega_{G1}^{G1}, \omega_{G2}^{G1}, Þ(\boldsymbol{G1}))$, where ꓘ is some capital used to produce *G1*, and by applying labor of S_{G1} and S_{G2} and paying them wages as in (I.1.h) above. This production function generates *G1*, that the market prices as worth Þ(G1). Similarly, for other entities.

j. **Entity $\boldsymbol{E_F}$ as say the Government is a Pure Pass Through.** In other words, if there is positive demand for asset *F*, the earnings from sale of F are distributed to the individuals as subsidies/cash transfer; if the demand for *F* is negative (i.e., investors need these assets to lever their portfolios), then it is financed through taxes. We will ignore taxes and subsidies/transfers and assume income is net of these issues as it makes the model cleaner with less variables. It could easily be added back on for a more complex model in the future.

2. **Asset Demand Set Up - Individual/Investor Consumption/Implicit Utility Function and Rates of Return on Assets**
   - **a.** Assume that the same two individuals, S_{G1} and S_{G2}, who supply labor and are compensated with wages (ω), seek to be able to satisfy their consumption function ala Modigliani's LCH (1952), as described in Modigliani-Brumberg (1980), subject to the budget constraint. The heterogenous investor model desired by Cochrane (2022) is that S_{G1} seeks cash flows *G1* and S_{G2} seeks cash flows *G2*, to satisfy their respective utility functions. Assume that the supply of labor is infinite for them to be able to purchase whatever they want, but a future variation of this model could assume limited labor supply so that goal achievement could be suppressed leading to constrained prices relative to a "free" equilibrium.
   - **b.** Both will seek to maximize goal-relative risk-adjusted performance, subject to an absolute and relative risk constraint (as discussed below) by creating portfolios that are linear combinations of *F*, *I*, *G1* and *G2*.
   - **c.** The solution to their optimization problem will give asset allocations to each of these assets for S_{G1} and S_{G1}. Once we have prices, $Þ(\boldsymbol{i})$, and the asset allocations for each, we can now estimate the cost of the portfolio for each investor/worker.
   - **d.** **The Implicit Utility Function of S_{G1} Investor**: $\boldsymbol{U_{SG1}}(G1, I, F, \omega_{G1}, \sigma_{G1}, \tau)$, subject to the constraint that total wages/income = cost of the optimal portfolio, and also subject to the absolute risk (σ_{G1}) and relative risk targets (τ) identified in their IPS, as shown to be the case for NMPERA and LACERA (or even SRA) earlier.
   - **e.** Therefore, for a GE to take place, income of S_{G1} and S_{G2} needed to buy the optimal portfolio must equal the wages they earn (to satisfy the budget constraint), and this allows us to impute an implicit production and labor allocation function that must satisfy this constraint.

All these key parameters can be summed up in Table 1.

CRITERIA	***E_{G1}***	***E_{G2}***	***E_I***	***E_F***	***S_{G1}***	***S_{G2}***
Maximize	Profit	Profit	Profit	Welfare	Utility	Utility
Earnings/ Receipts	Þ$(G1)$*Amount Sold	Þ$(G2)$*Amount Sold	Þ(I)*Amount Sold	Þ(F)*Amount Sold and Taxes	Wages (ω_{G1}) & Cash Transfers from E_F	Wages (ω_{G2}) & Cash Transfers from E_F
Payments/ Outflows	Wages (ω^{G1})	Wages (ω^{G2})	Wages (ω^{I})	Þ(F)*Amount Bought and Cash Transfers	Value of the Portfolio (& any taxes to fund govt lending)	Value of the Portfolio (and any taxes)
Primary Goal Assumption	Zero Economic Profit	Zero Economic Profit	Zero Economic Profit	Social Welfare	Maximize Relative to Goal Return	Maximize Relative to Goal Return
Solve For:	Þ$(G1)$	Þ$(G2)$	Þ(I)	Þ(F)	Asset Prices, allocation and total wages (ω_{G1})	Asset Prices, allocation and total wages (ω_{G1})

Table 1: Summary Parameters and Criteria in the GE model

Unlike the investment objectives, we do not have an equivalent and transparent function that we can articulate for how individuals allocate their labor (among the entities and leisure) or an exact production function, but we will demonstrate the conditions needed for any implicit function to satisfy in Section VI.

IV. Specific Assumptions for the GE Model based on the Stylized Economy

For convenience, capital letters will be used to represent entities, individuals and assets (with the exception of Tracking Error—TE), while Greek letters and lowercase letters will represent variables. The following assumptions are required for the model:

i. Assume a one-period setting.

ii. Investors/principals, S_{G1} and S_{G2}, each have a stochastic goal (in generic terms, *L* for "liability" or later, specifically *G1* and *G2*) and delegate to one (or more) agent(s) to achieve that goal.

iii. Principals seek the highest goal-relative, expected (net) risk-adjusted excess return. For simplicity, we exclude costs as that will be addressed in future research. If two portfolios have the same absolute and relative volatility (i.e., TE), they would prefer the portfolio with the higher expected risk-adjusted return. Risk is measured both as absolute volatility and relative risk (or tracking error). They have unlimited labor to offer to entities to earn sufficient income to purchase the optimal portfolio.

iv. Principals hire agents to take risk relative to their benchmark (tracking error) for one or more of three reasons: the principal is underfunded[15]; the principal is incapable of managing assets[16]; or agents claim to have skill in outperforming the benchmark. Principals permit limited relative risk budgets as they are uncertain about the skill of agents and can only observe past returns, not effort or skill. They extrapolate from past returns to forecast expected returns – a common industry practice.

v. Principals specify risk targets with a specific absolute risk level, equal to the risk (or volatility) of the relative safe asset (or GAD),[17] and a relative risk budget for their agents, TE(Target), or τ, based on their risk aversion. Notice that in this model (and unlike theory based on utility functions), "risk aversion" is very clearly identified and observed as shown earlier. Moreover, as Muralidhar (2000, 2001) shows, specifying purely an absolute risk target or just a tracking error target, and not both, will most likely lead to hiring a lucky agent (as an underspecified risk budget can be easily gamed by an intelligent agent) and shown below in Section V.

vi. There are two specific goals, *G1* and *G2*, respectively, each with representative investors (S_{G1} and S_{G2}, respectively) for that goal. Initially, assume that the

[15] In just the retirement world, Social Security systems, corporate and public defined benefit pension plans are underfunded and there is evidence that even individuals are not saving enough in their defined contribution plans. For example, US public pensions have a funded status on average < 80% with nearly $4trillion in AUM. Hence, this is a very reasonable assumption.

[16] In many pension funds, the board of the pension fund is composed of non-financial individuals who have other jobs. In the retail world, close to 40% of the population is financially unsophisticated.

[17] Modigliani and Modigliani (1997). Also, Jorion (2003) argues that normalizing absolute volatility, in exactly this manner, is the efficient way to budget tracking error risk.

representative investors are unable to trade with one another to establish the optimal general equilibrium equations for asset pricing. We term this goal-specific or "myopic" equilibrium with corresponding expected returns. Then, they are permitted to trade assets with each other to ensure a general equilibrium across all goals, eliminating any inconsistencies in pricing assets using the myopic approach.

vii. Both these goals have unique goal-replicating assets, *G1* and *G2*, respectively[18], with natural issuers as discussed earlier in Section II.

viii. Assume *F*, the traditional risk-free asset with zero volatility and zero correlation with all other assets, exists through its cash flow issuing entity, E_F.

ix. Also, there exists the generic risky asset, *I*, that we seek to price, issued through its cash flow issuing entity, E_I. *I* is different from *G1*, *G2* and *F,* and the absolute value of the correlation of *I* to *G1* and *G2* is different from unity.[19]

x. The supply of all assets is determined by the implicit production functions owned by the various entities. Where such assets do not exist, Section II has shown that they can be created with a bit of creativity and effort.

xi. Assume agents can invest in just a single risky asset to create the delegated portfolio relative to the goal. They invest either in *I* or the second goal-replicating asset. In principle, *I* could be a basket of risky assets. They can also invest in *F*. We do not put any restrictions on the sign of the allocations, except that the sum of allocations to the three assets equals 100%.

xii. Principals use M-cube risk-adjusted expected returns to evaluate agents. This is proved in Section V based on Muralidhar (2000) and Ambarish and Seigel (1996).

xiii. For both goals, TE(Target) or τ is identical. The TE(Target) or τ need not be identical as that also depends on the absolute volatility of the goal. This just simplifies the formulae, not the end result, and is easily relaxed later.

xiv. There are no transactions costs and unrestricted trading prevents arbitrage.

xv. There are equal proportions of investors for both goals (or with identical assets under management) so that the initial/simple model is free of the fraction of investors for

[18] Implicitly, this assumes that $\rho(G1, G2) \neq 1$ or -1.

[19] For simplicity, we are assuming that $\rho(I, G1)$ and $\rho(I, G2) \neq 1$ or -1. This constraint can be relaxed based on other correlations, but for now, it keeps the model simple.

each goal. Muralidhar (2023) provides the equilibrium equations when this is not true and it is briefly touched upon in Section VI (with the equation in a footnote).

If *I* is the single risky asset, L is the single stochastic goal, then the relative risk (also called Tracking Error or TE(*I, L*)) is defined as in equation (1), where σ(*) is the volatility and ρ(*) is the correlation between two assets.

$$\text{TE}(I, L) = \sqrt{[\sigma^2(I) - 2 * \rho(I,L) * \sigma(I) * \sigma(L) + \sigma^2(L)]} \qquad (1)$$

V. Postulates and Proofs

In this Section we lay out the model first and test it with real data from Brazil thereafter to see how the normative model works relative to actual data.

A. Solving the GRAPM Model to Derive Various Results.

We now proceed through eight postulates/proofs (and three figures) that lead us to the final conclusion that two goal-replicating assets or GADs and the absolute risk-free asset (*F*) are sufficient to ensure a GE, and allow for the pricing of all assets, that also satisfy the budget constraint of the individuals, and also allow for the specification of an implicit production compatible with such prices and wages (and the conditions that must be satisfied by an implicit labor allocation decision too). The detailed derivations are in the Appendices to make this easier to read and follow.

We first start with our most important issue, ignored in the agency literature; namely, to use a robust measure of skill. The measure of skill, that is robust and tenable in asset pricing theory is from Ambarish and Seigel (1996), as opposed to say Cornell (2009), and defined as below.

Assume *P* is a portfolio of an agent that is benchmarked to a generic goal, *L*, and *T* measures time or length of historical track record, and *r*(*) represent returns.

i. Skill is Critical in Agency Arrangements and Can Be Clearly Articulated

Ambarish and Seigel (1996) show that in order for skill to dominate noise, we need *T* (or number of data points) to satisfy the following equation, where *K* measures the degree of confidence (based on standard deviations of a unit normal; typically, 1, 2 or 3).[20] The proof is in Appendix 1.

$$T > \frac{K^2 * (\sigma_P^2 - \sigma_L^2 - 2 * \sigma_P * \sigma_L * \rho_{P,L})}{\left\{\left(\mu_P - \frac{\sigma_P^2}{2}\right) - \left(\mu_L - \frac{\sigma_L^2}{2}\right)\right\}^2} \tag{2a}$$

Muralidhar and U (1997) invert this equation and solve for *K* instead, since *T* is known when evaluating an agent, and this is shown in equation (2b). We term this "confidence in skill", when converted into percentage terms from the unit normal of *K*. This is clearer in Table 2 below.

$$\sqrt{T} * \left[\frac{\left\{\left(\mu_P - \frac{\sigma_P^2}{2}\right) - \left(\mu_L - \frac{\sigma_L^2}{2}\right)\right\}}{\sqrt{(\sigma_P^2 - \sigma_L^2 - 2 * \sigma_P * \sigma_L * \rho_{P,L})}}\right] > K \tag{2b}$$

One can easily see that the denominator is equation (1) and the numerator is not the simple difference between historical or expected returns as assumed in typical risk-adjusted performance measures like the Sharpe ratio (Sharpe 1994) or M-square (Modigliani-Modigliani 1997), but rather adjusted for half the variance of each of the respective return terms (so rather an Instantaneous Sharpe – Muralidhar 2015). This leads us to Postulate ii.

ii. *Where Skill is Critical (and decisions made relative to a stochastic goal), M-cube is the measure of risk-adjusted performance to be used.*

M-cube (Muralidhar 2000), an extension of the M-square of Modigliani and Modigliani (1997), demonstrates that typical risk-adjusted measures like the Sharpe ratio (Sharpe 1994) or the Differential Sharpe ratio, M-square or even GH1/GH2 (Graham and Harvey 1994, 1997) cannot rank agents identical to rankings based on measures of confidence in skill, as in

[20] Muralidhar and Seigel (2024) demonstrate that embedded in this formulation is also a notion of "implicit time", which is different from the number of data points calculation considered here.

equation (2b). This is a very important point as principals would ideally want only skillful agents or the arrangement cannot be incentive compatible.

What is M-cube? Modigliani-Modigliani (1997) make a critical point that risk-adjusted performance should not be measured as a ratio, but rather in return terms, making all comparisons equal by adjusting for risk (defined in their paper as absolute volatility as they base their measures, like Sharpe (1994), in an Expected Utility Theory/CAPM world, which interestingly has no delegation, and hence it is a bit asynchronous to examine risk-adjusted performance of agents using a model that ignores agency/delegation!). M-cube is represented in equation (3) and subject to 3 constraints highlighted in equations (4), (5) and (6). Stated otherwise, investors maximize expected, goal-relative, (net) risk-adjusted returns, as described in Section I given how investors specify objectives. Mathematically,

$$\text{Max E}[\text{r}(A) - \text{r}(L)], \tag{3}$$

where the risk-adjusted portfolio, A, is composed of the agent's portfolio, P, the goal hedging portfolio (or relative risk-free asset/GAD), L, and F (absolute risk-free asset) as in equation (4), subject to equations (5) and (6).

$$\text{E}[r(A)_{\text{P, L}|\tau}] = a^{P}_{\tau/L} \times \text{E}[r(P)] + l^{L}_{\tau/L} \times \text{E}[r(L)] + (1 - a^{P}_{\tau/L} - l^{L}_{\tau/L}) \times r(F) \tag{4}$$

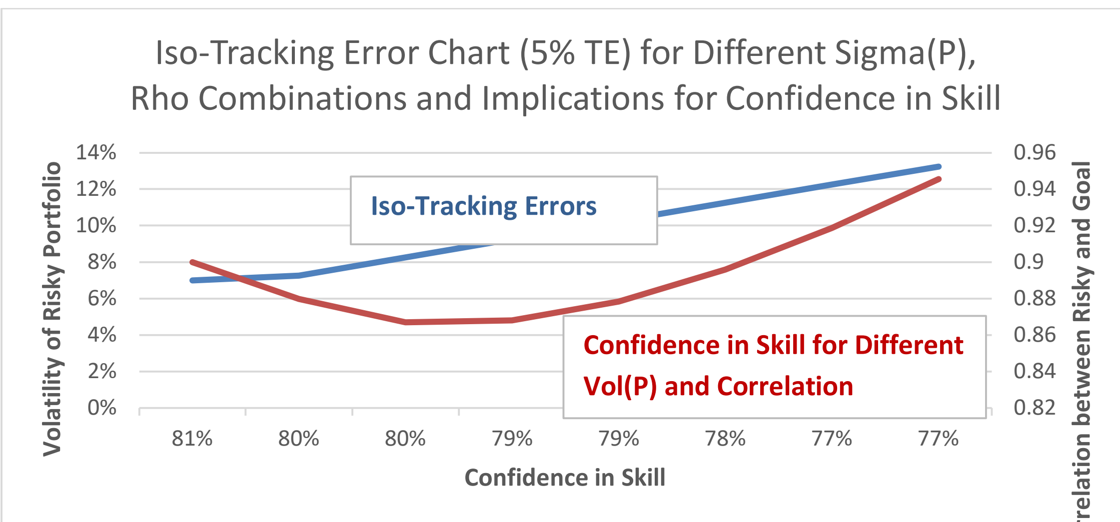
$$\text{s.t. } \sigma(A) = \sigma(L) \tag{5}$$

$$\text{and TE}(A, L) = \text{TE(Target}, L) \text{ or } \tau \tag{6}$$

where $a^{P}_{\tau/L}$ is the allocation to the risky (agent) portfolio, P, given the goal of L, and a target relative risk of τ; $l^{L}_{\tau/L}$ is the allocation to the goal-hedging/goal-replicating asset, L (and measures what is *de-facto* invested in the low-cost passive benchmark or "beta" in industry parlance). The super-script denotes the asset being allocated to; the subscript indicates the target relative risk (τ) and the goal-replicating asset (L). Similarly, $\text{E}[r(A)_{\text{x, y}|\tau}]$ is the expected return of A, assuming x is the risky asset, y is the goal, for τ target risk. The balance of the assets is invested in the traditional absolute risk-free asset, F (and measures leverage too).

This objective function is identical to Sharpe and Tint (1990), which shows that maximizing funded status (A/L) reduces to maximizing E[r(A) – r(L)], but Sharpe and Tint (1990) only focus on asset allocation and not on asset pricing or risk-adjusted performance. M-cube requires that all agents' risk-adjusted performance (r(A)s) have the same relative risk (TE) to L, but this must be achieved by ensuring that the risk-adjusted performance has been normalized to have the same volatility, $\sigma(L)$ as in M-square, AND target correlation, $\rho(\tau, L)$. This dual normalization is important because a specific TE value can be achieved by different combinations of portfolio volatility and correlation relative to the benchmark (equation 1). But each combination of volatility and correlation, assuming a fixed (historical or expected) outperformance by an agent, implies very different levels of confidence that the performance is skill-based as shown in equations (1) and (2b).

Figure 4: Iso-Tracking Error and Confidence in Skill for Various Volatility and Correlation Pairs



This is shown visually in Figure 4 (with data in Table 2), which plots Iso-Tracking errors (combinations of volatility of risky portfolio and correlations that give a 5% tracking error and plotted in blue), and the confidence in skill for these combinations (red line), assuming $\sigma(L)$ = 10%. Visually Figure 4, and numerically Table 2, show that if two agents have the same expected return and tracking error, but with different volatilities and correlations, then the principal is not agnostic between the two and cares deeply about how the tracking error has been generated as shown by the red line. They would ideally prefer to be on the highest point on the red line, but are limited by the constraint on the volatility of the risky portfolio (i.e., $\sigma(L)$ = 10%). Compare for example, portfolio A with H, in Table2, assuming the same excess return; both have the same tracking error (TE), but portfolio A, with a volatility of 7% and correlation of 0.9, has a confidence in skill of 81.3%; H with higher absolute volatility (13.25%) and correlation (0.945), has a lower confidence in skill of 76.6%. Hence, the dual normalization of M-cube demonstrates that it will dominate measures like the Sharpe ratio (used in Perold 2004 to establish the CAPM equilibrium) or M-square.

Portfolio	TE	Vol (L)	Vol (P)	Correlation	TE	Excess	K	Confidence
A	5%	10%	7%	0.9	5%	5%	0.887228832	81.3%
B	5%	10%	7.25%	0.879739741	5%	5%	0.847433474	80.2%
C	5%	10%	8.25%	0.867043912	5%	5%	0.831933266	79.7%
D	5%	10%	9.25%	0.867902587	5%	5%	0.814429008	79.2%
E	5%	10%	10.25%	0.878355844	5%	5%	0.794944625	78.7%
F	5%	10%	11.25%	0.895835855	5%	5%	0.773446276	78.0%
G	5%	10%	12.25%	0.918623405	5%	5%	0.749941014	77.3%
H	5%	10%	13.25%	0.945517778	5%	5%	0.724433315	76.6%

Table 2. Table to Demonstrate Different Combinations of Risky Portfolio Volatilities and Correlation can Deliver the Same Tracking Error (5%), but Different Confidence in Skill

iii. *M-cube provides very specific asset allocation recommendations to P, L and F.*

Modigliani-Modigliani (1997) demonstrate the dominance of their risk-adjusted ratio over say the Sharpe ratio (Sharpe 1994), as they provide the investor with valuable asset allocation recommendations. The derivation of M-cube is provided in Appendix II and the asset allocation results are as follows:

$$a^P_{\tau/L} = \frac{\sigma(L)}{\sigma(P)}\left[\sqrt{\frac{[1-\rho(\tau,L)^2]}{[1-\rho(P,L)^2]}}\right] = \frac{\sigma(L)}{\sigma(P)}\left[\frac{\varphi(\tau,L)}{\varphi(P,L)}\right], \text{ where } \varphi(I,J) = \sqrt{[1-\rho(I,J)^2]} \tag{7}$$

$$l^L_{\tau/L} = \rho(\tau,L) - \rho(P,L) \times \left[\sqrt{\frac{[1-\rho(\tau,L)^2]}{[1-\rho(P,L)^2]}}\right] = \rho(\tau,L) - \rho(P,L) \times \left[\frac{\varphi(\tau,L)}{\varphi(P,L)}\right] \tag{8}$$

Which then gives the allocation to F. In industry parlance, l measures the beta (or benchmark allocation implicit in the portfolio, which is cost-lessly achieved), and ($1\text{-}a\text{-}l$) measures leverage (again, which is costless and should be compensated). In effect, M-cube describes the true alpha, for a given absolute and relative risk budget, after normalizing for leverage and beta, which principals should not pay for. This is another reason why this measure dominates many others currently used in practice as they do not normalize for beta.

iv. *M-cube is a de-facto demand function for all assets and allocations are "view neutral"*

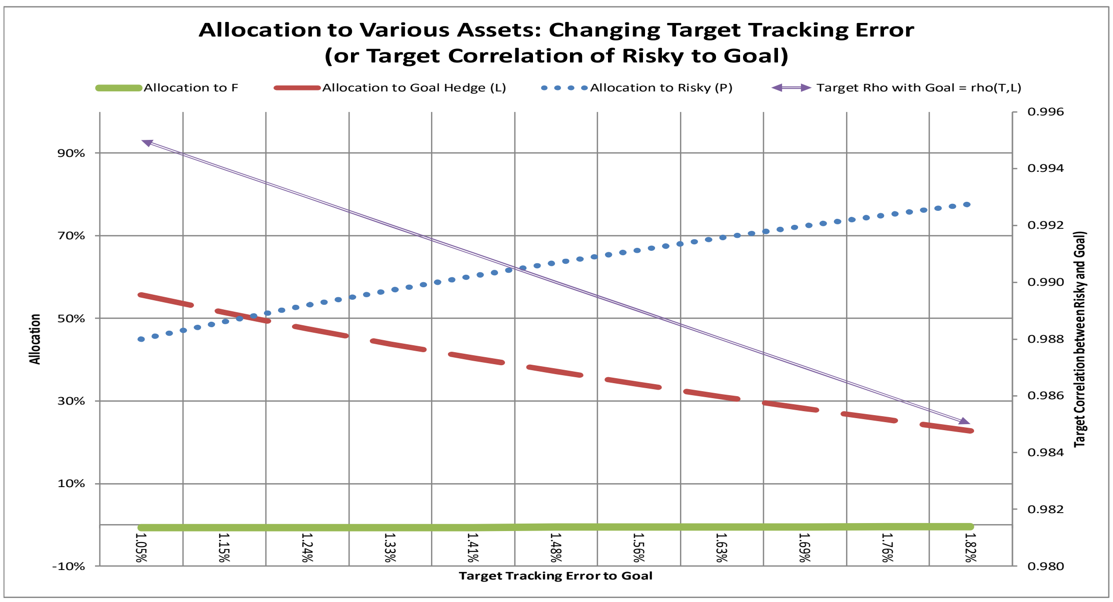
The most important observation from (7) and (8) is that there is no expected return term in either a and l and hence the "view neutral" claim, unlike say traditional Markowitz (1952), which require highly imprecise expected return forecasts. Markowitz (1952) notes clearly that his model depends on "beliefs", but adds in footnote 7, "[T]his paper does not consider the difficult question of how investors do (or should) form their probability beliefs," and this lack of clarity and difficulty has plagued institutional and retail investing for decades.

If $\rho(P,L)$ is the correlation of returns of L and P of each agent (which is easily calculated from actual data), and $\rho(\tau,L)$ is the target correlation[21], which is specified by the principal, we can solve for the optimal asset allocation "a" as in equation (7), and "l" in equation (8). From (7), the allocation to risky assets is an increasing, linear function of the volatility of the goal, a decreasing nonlinear function of the target correlation (i.e., risk aversion), and increasing, nonlinear function to the correlation of the asset to the goal. Conversely, it is a decreasing function of its own volatility. Note that in the M-cube approach, the investor decides which asset(s) form the risky portfolio. This freedom to choose what is "risky" is used to derive GRAPM and reflects reality. Most, if not all investors do not perform Two Fund Separation of Tobin (1958). For example, one fund may decide that private credit is the risky asset they want to allocate to, and the second fund may decide that private equity is the risky asset of choice as there is no unique "market portfolio" in practice as established/required in CAPM.

[21] Which is an exogenous relative risk target value specified by the principal and derived from TE(Target) once $\sigma(L)$ is known.

Similarly, from (8), the allocation to the goal hedging asset is related positively and non-linearly to the target correlation, and negatively and non-linearly to the correlation of the risky asset to the goal. The allocation to the goal-replicating asset is independent of any volatility terms. These results are intuitively obvious as the allocation to risky assets should increase with increases in absolute and relative risk targets. Similarly, the allocation to the goal hedging asset will increase, the lower the relative risk target. Figure 5 provides a visual of how one might establish a demand curve for these assets, based on this explicit measure of risk budget.

Figure 5: Allocation to Risky Assets, Goal-Replicating Asset, and Absolute Risk-Free Asset for different Target Tracking Errors (by Varying Correlations).



Assume reasonable values of σ(*L*) = 10.5%, σ(*P*) = 10.5% and ρ(*P,L*) = 0.99. Then a TE(Target) = 1.5%, implies that ρ(*τ,L*) ≈ 0.995. Figure 5 provides a range of tracking error targets around this TE(Target) by just varying ρ(*τ,L*). For low levels of relative risk (i.e., funded status or relative risk aversion is high), allocation to the liability hedge is high (dashed line) and declines thereafter as the allocation to risky assets (dotted line) rises.[22] The double-lined arrow provides the target correlation, ρ(*T,L*), for these TE(Target) levels. The allocation

[22] In this setting, the non-linearity is not as obvious as in the example in Muralidhar and Shin (2013) that examines likely allocations for a corporate pension fund, and hence a different goal-replicating asset (as corporate pension funds usually discount liabilities with a bond ladder with a duration of approximately 10 to 15 years depending on the maturity of the plan).

to *F* ranges from -1% to 0% for these settings. In the example provided in Muralidhar and Shin (2013), they are substantially different from zero. Notice, these demand curves, charted in Figure 5, satisfy Cassel (1924) statement of criteria for his GE model!

Interestingly, "*a*" and "*l*" will feature in the asset pricing equation as a way of capturing the dual value of an asset in serving as a risky asset for some goals, and hedging some part of other goals, respectively.

v. *In GRAPM, there are 3 classes of assets*

Implicit in the discussion up until now, but critical for the GE claim and the asset pricing model, is that there are three classes of assets: (a) a single absolute risk-free asset, *F*; (b) multiple relative risk-free assets (G_i); and (c) All other (risky) assets, such as *I*. GRAPM provides unique asset pricing formulae for each of these assets.

vi. *A Myopic Equilibrium in G1 (using G2 and I as risky assets) gives us one equilibrium equation*

Using Assumption vi, we can create two risk-adjusted portfolios relative to the *G1* goal, both with identical volatility and correlation to *G1 (*say Brazil's RendA+ retirement bond); one using *G2* (say Brazil's Educa+ education bond) as the risky asset and the other using *I* (say a single stock or the entire index like the S&P500 Equity Index in Brazilian Real terms) as the risky asset. While *I* and *G2* are unique, their risk-adjusted returns, created by a linear combination of that asset, *G1* and *F*, must be identical otherwise there would be zero demand for the asset with the lower risk-adjusted return (and those assets would cease to exist leaving only assets that satisfy this condition) - Assumption iii. This gives us equations (9a, 9b and 10a and 10b), derived in Appendix III, and visualized in the top half of Figure 6.

If $a'_{G1} = \frac{\sigma(I)}{\sigma(G2)}\left[\frac{\varphi(I,G1)}{\varphi(G2,G1)}\right]$ (9a)

And $l'_{G1} = \frac{\sigma(I)}{\sigma(G1)} \times \rho(I,G1) - \frac{\sigma(I)}{\sigma(G1)} \times \rho(G2,G1) \times \left[\frac{\varphi(I,G1)}{\varphi(G2,G1)}\right]$ (9b)

Define $E[r(I)_{G1}]$ as the expected return of asset *I* in *G1*.

$$E[r(I)_{G1}] = a'_{G1} \times E[r(G2)] + l'_{G1} \times E[r(G1)] + (1 - a'_{G1} - l'_{G1}) \times r(F) \quad (10a)$$

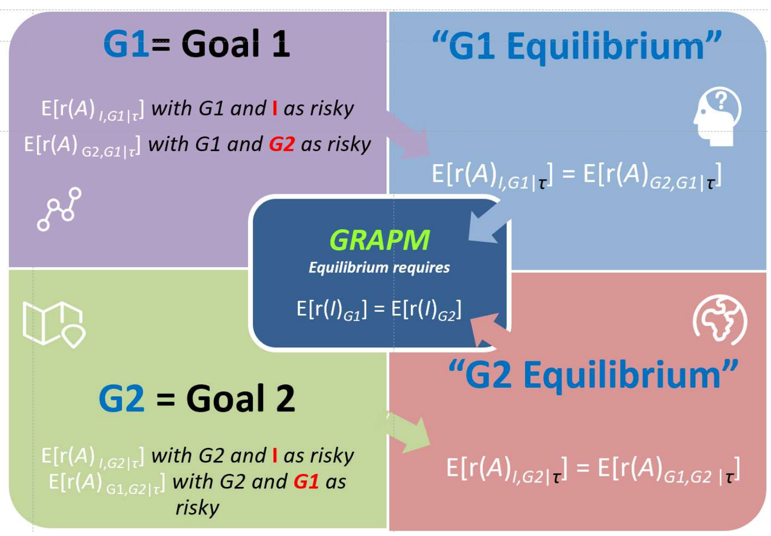
Alternatively, equation (10a) can be expressed in the following manner by rearranging terms.

$$E[r(I)_{G1} - r(F)] = a'_{G1} \times E[r(G2) - r(F)] + l'_{G1} \times E[r(G1) - r(F)] \quad (10b)$$

vii. *A Myopic Equilibrium in G2 (using G1 and I as risky assets) gives the second equilibrium equation*

Using Assumptions vi and iii, we repeat the process to derive the equation for *I* for *G2* investors (and also provided in Appendix III) and visualized in the bottom half of Figure 6. One can see this as a much more reliable heterogenous investor model based on goals and portfolio construction, and satisfying Cochrane (2022), rather than complex utility function specifications as in Chan and Kogan (2002).

Figure 6: Visualizing the Myopic G1 (top half) and G2 (bottom half) Equilibrium and then the Global/General (center box) Equilibrium.



If $a''_{G2} = \frac{\sigma(I)}{\sigma(G1)} \left[\frac{\varphi(I,G2)}{\varphi(G1,G2)}\right]$ (11a)

And $l''_{G2} = \frac{\sigma(I)}{\sigma(G2)} \times \rho(I,G2) - \frac{\sigma(I)}{\sigma(G2)} \times \rho(G1,G2) \times \left[\frac{\varphi(I,G2)}{\varphi(G1,G2)}\right]$ (11b)

Then

$$E[r(I)_{G2}] = a''_{G2} \times E[r(G1)] + l''_{G2} \times E[r(G2)] + (1 - a''_{G2} - l''_{G2}) \times r(F) \qquad (12a)$$

Or alternatively, rearranging terms in equation (12a),

$$E[r(I)_{G2} - r(F)] = a''_{G2} \times E[r(G1) - r(F)] + l''_{G2} \times E[r(G2) - r(F)] \qquad (12b)$$

viii. *Allowing free trade ensures a Janus Equilibrium with clear formulae for the expected returns of all three classes of assets*

Again, the proof is provided in Appendix III and visualized in the box in the center of Figure 6 in terms of process. If we define

$$X_{I,G1,G2} = \varphi(I,G1) - \rho(I,G2) \times \varphi(I,G1) + \rho(G1,G2) \times \varphi(I,G2) \qquad (13)$$

$$Y_{I,G1,G2} = \varphi(I,G2) - \rho(I,G1) \times \varphi(I,G2) + \rho(G1,G2) \times \varphi(I,G1) \qquad (14)$$

Then for the absolute risk-free asset,

$$r(F)_{G1,G2} = \frac{\left\{E[r(G1) - \frac{\sigma(G1)}{\sigma(G2)} \times (\frac{X_{I,G1,G2}}{Y_{I,G1,G2}}) \times E[r(G2)]\right\}}{\left[1 - \frac{\sigma(G1)}{\sigma(G2)} \times (\frac{X_{I,G1,G2}}{Y_{I,G1,G2}})\right]} \qquad (15)$$

Note that $r(F)_{G2,G1}$ while different in terms of formula, must give the same value (and we show this in the numerical example below using actual Brazilian data).

For the goal-replicating assets, equation (16) must hold,

$$E[r(G1) - r(F)] = \frac{\sigma(G1)}{\sigma(G2)} \times (\frac{X_{I,G1,G2}}{Y_{I,G1,G2}}) \times E[r(G2) - r(F)] \qquad (16)$$

And finally, for all risk assets, the following equation would apply:

$$E[r(I) - r(F)] = Z_{I,G1/G2} \times E[r(G1) - r(F)] \qquad (17)$$

Where Zeta is a form of a "covariance term" as beta is in CAPM, defined as

$$Z_{I,G1/G2} = \frac{\sigma(I)}{\sigma(G1)} \times \{\left[\frac{\varphi(I,G2)}{\varphi(G1,G2)}\right] + (\frac{Y_{I,G1,G2}}{X_{I,G1,G2}}) \times \rho(I,G2) - (\frac{Y_{I,G1,G2}}{X_{I,G1,G2}}) \times \rho(G1,G2) \times \left[\frac{\varphi(I,G2)}{\varphi(G1,G2)}\right]\} \quad (18)$$

Notice, that Zeta is a function of two volatilities, $\sigma(I)$ and $\sigma(G1)$, (as in CAPM), but three correlations: $\rho(G1,G2)$, $\rho(I,G2)$ and $\rho(I,G1)$. In CAPM, there is just a single correlation because CAPM is a highly specific/restrictive model that ignores the realities and relative decision-making of practice. One other key point to note, equation (17) cannot be used for assets that are goal-replicating; instead, equation (16) must be used for goal-replicating assets. Further, since X and Y are functions of $\varphi(I,J)$, Zeta is not defined when $\rho(G1,G2)$, $\rho(I,G2)$ or $\rho(I,G1) = 1$ (which would violate the uniqueness assumption), unless the other correlations are = 0. These extreme cases are examined in Muralidhar (2024b).

Interestingly, Zeta can be restated in terms of "*a*" and "*l*" as follows and has been interpreted in Muralidhar (2023) to show that the value of an asset is its role as a goal hedging asset, but also potentially as a risky asset for other goals.

$$Z_{I,G1/G2} = \frac{a^{G1}_{T/G2}}{a^{I}_{T/G2}} + \frac{\sigma(I)}{\sigma(G1)} \times (\frac{Y_{I,G1,G2}}{X_{I,G1,G2}}) \times l^{G2}_{I/G2} \quad (19)$$

Similarly, express E[r(*I*)] as a function of E[r(*G2*)] as the other risky asset, r(*F*), volatilities and correlations (assuming *G1* is the goal) as in equation (20).

$$E[r(I) - r(F)] = Z_{I,G2/G1} \times E[r(G2) - r(F)] \quad (20)$$

We can see these dual equations in (17) and (20) for *I* as a pair-wise or "Janus" equilibrium.[23] Interestingly, given equations (20), (16) and (17), there is a unique relationship between $Z_{I,G1/G2}$ and $Z_{I,G2/G1}$; namely, that

$$Z_{I,G2/G1} = \frac{\sigma(G1)}{\sigma(G2)} \times (\frac{X_{I,G1,G2}}{Y_{I,G1,G2}}) \times Z_{I,G1/G2} \quad (21)$$

[23] In this example, the Janus duality is demonstrated by the fact that *I* can be priced in this pair-wise equilibrium with *G1* as risky (*G2* as goal) or vice versa. This duality is unchanged even when we add more goal-replicating assets, as we only use two at a time to create the asset pricing equation for *I*.

This equilibrium condition implies that the 3 correlations in the 2 goal-replicating asset case are "entangled" in that they cannot take any arbitrary value (and this will make it hard to test this normative model despite Brazil issuing such goal-replicating assets, until all investors adopt GRAPM). This will be critical when many more goal-replicating assets are added (as *I* was a generic single asset or a portfolio of assets, and hence adding more *I*s is more complex and covered in Muralidhar 2023 and touched upon briefly in Section VI).

B. Testing the Model with Live Data

To show how this model works and satisfies the claim that we can price any assets with just two GADs and the absolute risk-free asset, we assume a relatively simple case of two goal replicating assets, $G1$ and $G2$, with given volatilities, and correlations, derived from actual data from Brazil from August 1, 2023 – November 29, 2024. However, recall that GRAPM is a normative model, and hence we show the historical values versus what the GE model would require if all investors adopted a GRAPM approach, using a two GADs case. Appendix VII highlights the 3 GADs case.

We use real data to establish correlations and volatilities of RendA+ 2060 (starts paying in 2060) and Educa+2030 (starts paying in 2030) visualized in Figures 3A and 3B. The risky asset is the S&P500 US Equity Index in Brazilian Real. The key statistics on these assets is shown in Table 3. The historical (arithmetic) annualized returns, assuming a 260-day year, are also provided for completeness. We are solving for future expected returns/asset prices. For completeness, we also provide the data on alternate Educa+ (2040) and RendA+ (2035) as Brazil created a series of these bonds for various start and end dates of cash flows for all age groups and hence the pricing model can be tested on any number of combinations of two or more GADs (and we use RendA+ 2035 in Appendix VII for the 3 GADs example). The key inputs are highlighted in green. We assumed a 3% Target TE and used actual values of $\sigma(G1) = 35.98\%$, $\sigma(G2) = 7.13\%$ and $\sigma(I) = 13.49\%$. The relevant correlations are Educa 2030-RendA 2060 (0.873), Educa2030-S&P500BRL (-0.142) and RendA2060-S&P500BRL (-0.162).

	Educa 2030	Educa 2040	RendA 2035	RendA 2060	S&P500 TR ($)	BRL	S&P500 (BRL)
Annualized Return	0.25%	-6.64%	-7.41%	-21.67%	22.45%	15.93%	38.38%
Annualized Risk	7.13%	14.45%	14.88%	35.98%	12.55%	11.30%	13.49%
Return/Risk	0.04	-0.46	-0.50	-0.60	1.79	1.41	2.85
	Correlation Matrix						
	Educa 2030	**Educa 2040**	**RendA 2035**	**RendA 2060**	**S&P500 TR ($)**	**BRL**	**S&P500 (BRL)**
Educa 2030	1.000	0.938	0.936	0.873	0.018	-0.190	-0.142
Educa 2040		1.000	0.988	0.923	0.007	-0.186	-0.149
RendA 2035			1.000	0.934	0.010	-0.188	-0.148
RendA 2060				1.000	0.000	-0.193	-0.162
S&P500 TR ($)					1.000	-0.362	0.626
BRL						1.000	0.500
S&P500 (BRL)							1.000

Table 3: Raw Data on Various Educa+ and RendA+ Bonds, S&P500 and the Brazilian Real (Aug 1, 2023 = Nov 29, 2024).

i. The Asset Pricing Results and Comparison to GE Values

We solve for r(*F*), E[r(*G1*)], E[r(*G2*)] and E[r(*I*)], as shown in Table 4a below. The data highlighted in yellow are the equilibrium expected (forward-looking) returns for each of the assets, given correlation and absolute and relative volatility assumptions. The bottom part of Table 4a also shows the implications of a fixed target tracking error and different volatilities on the target correlation. Assuming that the price of each asset $j = \frac{1}{(1+\mathrm{E}[rj])}$, since this is a one period model, we also include the implied price of each asset in the last column, and highlighted in blue. Interestingly, while the model solves out (and does not converge to a solution with a 2% tracking error), it also does not converge to a clean GE equilibrium solution unless we assume that the RendA2060-S&P500BRL correlation is -0.1417 as well, slightly different from reality. This is shown in the table with a dark orange highlight in the correlation section. The returns from the model are very reasonable and not outlandish, and the lower duration assets, Educa2030, has a lower expected return than RendA2060. Notice that this asset pricing model is different from the current asset pricing model offered by Tesouro Nacional that is used to price (and bid-ask spreads three times a day) these instruments daily based on an implied real yield curve.

	ASSET PRICING								
	Expected Return	Volatility	Return/Risk	Correlation				Price
Asset				F	S&P500 TR	RENDA 2060	EDUCA 2030	
	2	3	4	5	6	7	8	
F	2.02%	0		1	0	0	0	0.9802
S&P500 TR	8.29%	13.5%	0.61	0	1	-0.1417	-0.142	0.9234
RENDA 2060	4.72%	36.0%	0.13	0		1	0.873	0.9550
EDUCA 2030	2.55%	7.1%	0.36	0			1	0.9751
Target Tracking Error		3%						
ρ(τ,G1)		0.997						
ρ(τ,G2)		0.911						

r(I) - r(F)	6.27%
r(G1) - r(F)	2.70%
r(G2) - r(F)	0.54%

Table 4a. The 2 GADs Case with 3% Target Tracking Error and Solving for r(*F*), r(*RENDA 2060*), r(*EDUCA 2030*), and S&P500 in BRL terms.

In Table 4b, we demonstrate that the equilibrium conditions are achieved given how many checks and balances exist in this model and there are no free parameters. We need (a) the absolute risk-free asset, *F*, to be identical regardless of which asset is used as primary (top panel); (b) consistency between r(*G1*) and r(*G2*) as relative risk-free assets (middle panel); and (c) the pricing of *I* (equation 21, tested with real data in the bottom panel). This also shows there are no free parameters removing the CAPM critique of John Cochrane.

	Using G1 as primary	*Using G2 as primary*
r(F)	2.02%	2.02%

Test of Expected Returns	r(G1) - r(F) LHS	r(G1) - r(F) RHS	Equilibrium
r(G1) - r(F) = [{σ(G1)/σ(G2)}*{X/Y}*{r(G2) - r(F)}]	2.70%	2.70%	-0.007%

Z(I, G2 = risky; G1 = goal)	0.2191
{σ(G1)/σ(G2)}	5.05
{X/Y}	1.0002
Z(I, G1 = risky; G2 = goal)	0.0435
Test: Is (Z(I, G2, G1) = {σ(G1)/σ(G2)}*({X/Y}* Z(I,G1, G2)	-0.001

Table 4b: Confirming Key Equilibrium Conditions for all Three Classes of Assets

ii. Asset Allocation and the Cost of the Portfolio

For completeness, asset allocations to various assets for *G1* and *G2* investors are provided in Table 4c, but before we do so, we bring back the wage/budget constraint equations. These asset prices and wages must satisfy the production function for all three profit-seeking entities, as we assumed that E_F is a pass through. The wages earned by E_{G1} are shown in equation 22 and E_{G2} are shown in equation 23, but will need to be sufficient to finance the portfolio.

$$\omega_{G1} = \omega_{G1}^{G1} + \omega_{G1}^{G2} + \omega_{G1}^{I} = 0.9420 \quad (22)$$

$$\omega_{G2} = \omega_{G2}^{G1} + \omega_{G2}^{G2} + \omega_{G2}^{I} = 0.9628 \quad (23)$$

We also report the total cost of both portfolios as that will also need to be the total wages paid by the entities that earn zero economic profit.

	RENDA 2060		EDUCA 2030		
	ASSET ALLOCATION				
		Allocation		Allocation	
	Asset		Asset		
	F	-23.29%	F	-19.01%	
	S&P500 TR	22.45%	S&P500 TR	21.97%	
	RENDA 2060	100.84%	EDUCA 2030	97.04%	
	Total	100.00%	Total	100.00%	
Portfolio Price or Wage for Investor		0.9420		0.9628	1.9048

Table 4c: Asset Allocation for Both Investors and Total Cost of Both Portfolios

The allocations look somewhat similar to *I* for the two different goals, and the allocation to *F* is negative (i.e., leverage) for both goals as well. But that could be because the two goals are highly correlated. This will be tested in future research. If we used the other GAD as the risky asset, obviously the portfolio allocations will be different as will the portfolio price (in this case, higher for both goals).

iii. Wages Earned

The total funds raised by all four entities are reported in Table 4d as well as the likely wages paid to employees (absent a clearly specified production function). In other words, the implicit production function will need to satisfy these constraints. By assumption, the "wages paid" by E_F is effectively the tax imposed on citizens to allow the entity to lend to investors to achieve their goals, but at a market rate of return. The various equilibrium conditions we can specify are provided in Appendix VI.

	Asset Sale Value $	Total Assets Sold	Wages Paid to Workers
F	-0.4147	-42.30%	**-0.4147**
S&P500 TR	0.4102	44.42%	0.4102
RENDA 2060	0.9630	100.84%	0.9630
EDUCA 2030	0.9463	97.04%	0.9463
	1.9048	2.0000	1.9048

Table 4d. Total Assets Sold and Value Raised by each Entity for Sale of Cash Flows based on Equilibrium Prices (and Implicit Production Function and Wage Allocation)

Table 4d is interesting as in this simple case, the need for *G1* and *G2* investors to leverage implies that the government instead of issuing T-bills, has effectively injected liquidity into the market which is not unrealistic (e.g., the Federal Reserve Bank has introduced $6.9 trillion into the market as of December 2024 by buying up a lot of junk assets from weak financial institutions). We can assume the government, unlike for-profit entities, has a welfare function to make the economy function effectively. The sum/net total of assets equals 200% and each entity has either a positive or negative inflow of cash to meet their financing needs for labor, capital (which we ignored for simplicity) and potentially economic profit. We do not need to make a hard assumption about perfect competition as was the case with the AD GE model.

One possible work allocation decision could be that S_{Gi} (S_{Gj}) earns the wage paid by E_{Gj} (E_{Gi}), plus one half of the receipts/payments for F (i.e., split the burden equally among both individuals), with the balance earned through work for E_I. This is shown in Table 4 e. We could do the same if we used the other GAD as the risky asset.

COST		WAGE	OTHER GOAL	HALF TAX /F	I WAGE	TOTAL WAGE
SG1 COST	0.9420	***SG1 WAGE***	0.9463	-0.207	0.2028	0.9420
SG2 COST	0.9628	***SG2 WAGE***	0.9630	-0.207	0.2068	0.9628
TOTAL	1.9048	***TOTAL***	1.9093	-0.4140	0.4095	1.9048

Table 4e. One Possible Work Allocation Function Results

iv. Production Functions, Other Issues and Adding N Goal-Replicating Assets

And hence $\emptyset_{\boldsymbol{G1}}(G1, \text{⅄}, \omega^{G1}_{G1}, \omega^{G1}_{G2}, \text{Þ}(\boldsymbol{G1}))$, $\emptyset_{\boldsymbol{G2}}$, and $\emptyset_{\boldsymbol{I}}$ must be the production functions that satisfies the solution to these prices of cash flows, total issuance, and wages paid to each worker. This all gives us a final GE condition (shown in Appendix VI), starting from the wage equations (22 and 23), that the

Budget constraints of both investors (LHS of the equation) = Total Revenue Generated by Sale of Shares by the three entities (RHS of the equation).

This seems trivial and not worth mentioning, except that we require more entities than investors, as we need a single risky asset (and absolute risk-free asset, F). We show this in equation (24).

Value of Portfolio *G1* + Value of Portfolio *G2* = $\text{Þ}(G1)$*Amount of Shares Sold (*G1*) + $\text{Þ}(G2)$*Amount of Shares Sold (*G2*) + $\text{Þ}(I)$*Amount of Shares Sold (*I*) (24)

Muralidhar (2023, 2024a) provides a numerical example with 3 GADs and the set of equations for the three GAD case is shown in Appendix IV and a numerical example of adding real (and modified) data if say Renda+ 2035 is added is provided in Appendix VII. The key takeaway from Appendix VII is that even though you get expected returns/prices/wages with just 2 GADs, these equilibrium values do change meaningfully if you add more GADs and hence one may never know the true forecasts if we cannot classify all assets into GAD or "risky". Regardless, for completeness, if we extend it to N goals/GADs, then the following must hold:

i. For each risky asset I, if there are N goal-replicating assets, then there will be N x (N-1) equations for I.
ii. There will be {N x (N-1)}/2 equilibrium equations for goal-replicating assets.
iii. For each I, there will be N equalities for Z.
iv. For each I, there will be N-1 equations that express the equilibrium between the X and Y variables.
v. There are also N equations for F.

VI. Comparison with Arrow-Debreu Model/Securities and Extensions

We would strongly disagree with Cochrane (2022) claims about a GE model; namely, that GE starts with a deep, powerful, and frequently overlooked theorem: The average investor must hold the market portfolio; Any deviation from the market portfolio is a financial zero-sum game; Portfolio theory must be all about heterogeneity; and portfolio formation begins with risk management. While we strongly agree with the last two points, and have shown how heterogeneity of cash flows/goals, and clear risk management objectives drive the GRAPM model, in our GE/GRAPM model, and in reality, there is no "market portfolio"! And given the fact that some investors may have myopic perspectives as is the case of say the 5 public pension plans under the purview of the New York City pension system and the SAME investment team (but each with their own forecasts of expected returns, correlations and volatilities for a range of complex reasons), there could be a solution other than a zero sum game if some investors are willing to be freed from regulatory and other constraints to have a global versus myopic perspective.

a. Comparison with AD Model

This model clearly satisfies many of the conditions of Cassel (1924), Wald (1951), the original AD (1954) model and even Cochrane (2022). Table 5 offers a comparison between the two approaches (and most of the other GE models are not too dissimilar as the assumptions are largely divorced from reality, including the ICAPM, CIR, Lucas model or the Gomes model as discussed in the Introduction). The simple AD world is a one-period world like the assumption in this paper and a clear extension would be to assume a stochastic setting so that time is a critical variable, especially given its role in the skill equation. Additionally, AD focused on the "basic economic motivation in the choice of a consumption vector", and GRAPM GE model is based on GADs to satisfy these goal-based cash flows to satisfy specific consumption (vector) needs. However, the equilibrium in the AD world, is based on complete markets and the state space, *S,* being completely spanned with as many AD securities. The GRAPM model requires just two GADs and adding more securities just create a lattice of equilibrium equations. Further, the AD model assumes that all payoffs are linearly independent (or that they are uncorrelated and obviously not perfectly correlated); the GAD model only requires the latter constraint; i.e., that assets are unique (as was shown to be case in reality as well in Table 4a). Hence it is a much more flexible/less rigid model.

Moreover, AD assumes a utility function and risk aversion neither of which are observed in practice, where the GRAPM model is based on strictly observable and controllable parameters, based entirely on investment practice, and could be complemented with assumed production and work allocation functions, but they are not critical to the GE solutions. Finally, as noted in the Introduction, the AD assumptions lead to interesting insights about prices in general equilibrium, but there is no practical relevance, whereas GRAPM provides asset pricing implications for three classes of assets, as well as asset allocation that is view neutral (which the AD model does not even address), and based on a consistent risk-adjusted performance measure that accounts for stochastic liabilities and ranks agents consistent with measures of confidence in skill (again, not considered in the AD model). There are no free parameters in the GRAPM model as demonstrated earlier and also in Muralidhar (2024b), especially in the last chapter which uses a vector approach to prove the "no free parameter" claim.

Issue	Arrow-Debreu	Goals-based AD
Payoff	1 or 0 depending on "state". **ORTHOGONAL**	Series of cash flows to match goal. Only condition = **UNIQUE**
How Many	Spanning = Unknowable	Only # of Goals, but can price all assets with just 2 GADS and F
Issuer	None – **cannot exist**	Governments, schools, pharma companies - **Exists**
Asset Pricing	Yes – easy (**Cochrane - Free Parameter problem**)	Yes – also price F (**no free parameters**)
Asset Allocation	Silent	**Yes – Goal-consistent**
Defined	**No** – states are arbitrary	**Yes** – goals very specific (retirement, education etc)
Capture Reality	**No** - Pfleiderer Chameleon	Yes

Table 5. Comparing the Arrow-Debreu Model to the GADs Model

b. Extensions

In addition to the extension in continuous time, Muralidhar (2023) considers the following extensions:

(a) Different weights of investor types: Adding the weights of each type of investor in the economy.[24] If we include this assumption, Zeta becomes a function of the weights of each goal based on the assets allocated to the goal (say w_{G1} and w_{G2}, respectively); more explicitly, $X_{I,G1,G2}$ and $Y_{I,G1,G2}$ in equations (13) and (14) include w_{G1} and w_{G2}, but the allocation formulae are unchanged.[25];

(b) Multiple agents or multiple asset portfolios: Assume the agent's portfolio, *I*, is made up of multiple assets as opposed to a single risky (or alternatively, that an investor hires

[24] Thanks to Prof. Robert C. Merton for stimulating this insight.

[25] If w_{G1} is the weight of *G1* and w_{G2} is the weight of *G2* (such that the sum is 1), $X_{I,G1,G2}^{w_{G1},w_{G2}} = w_{G1} \times \varphi(I,G1) - w_{G2} \times \rho(I,G2) \times \varphi(I,G1) + w_{G2} \times \rho(G1,G2) \times \varphi(I,G2)$ and $Y_{I,G1,G2}^{w_{G1},w_{G2}} = w_{G2} \times \varphi(I,G2) - w_{G1} \times \rho(I,G1) \times \varphi(I,G2) + w_{G1} \times \rho(G1,G2) \times \varphi(I,G1)$.

multiple agents to diversify the risk of being wrong). This just makes the model a bit more complicated. Because the correlation of a portfolio of assets to a goal-specific asset is nothing more than a weighted sum of each asset correlation to the goal-specific asset, the more complex model can be solved using M-cube risk-adjusted performance. Muralidhar (2001) demonstrates how this is achieved (in the context of hiring multiple agents) and further how assets that might have been considered valuable from a diversification perspective in an absolute return-risk world, may be sub-optimal in a relative risk world (and vice versa).[26] If M is the portfolio of risky assets, such that

$$r(M) = \sum_j w_j r(j), \text{ then } \rho(M,G1) = \frac{\sum_j w_j \rho(j,G1)\sigma(j)}{\sigma(M)}. \quad (27)$$

In this multi-asset (or multi-agent) setting, the allocation to each risky asset j that is in this portfolio $M = a^*w_j$ and can be solved iteratively (where "a" is the allocation to the risky asset from the M-cube solution). This potentially leads to an interesting new research avenue as to how the optimal (multi-asset) risky portfolio in *G1* relates to the optimal risky portfolio in *G2*. This offers a new twist to the notion of diversification and also has some ability to explain why many pension funds experienced declines in funded status during the equity crises of 2000-2 and 2008. In short, these pensions used mean-variance optimization (in an absolute return-risk space) to establish optimal portfolios (implicitly assuming as in MPT/CAPM that the goal is deterministic and maximizing wealth), while ignoring the fact that they had a stochastic goal that essentially resembled a long-duration bond asset. Hence, assets that looked attractive from a diversification perspective in MPT (e.g., equities) proved to be highly risky and even negatively correlated to the goal during crises. These assets might not have been included in a GRAPM portfolio for a limited relative risk budget. Coqueret et al (2017) attempt to address this challenge of capturing the goal-hedging value of specific equities as opposed to holding a "market portfolio" proxy. One can also potentially see this from Figure 1.

[26] Muralidhar (2001) goes further to demonstrate how this approach is superior to naïve maximization of differential Sharpe ratios (also known as Information ratios) and other techniques that do not rank multiple agent portfolios consistent with measures of confidence in skill.

(c) Different Risk Budgets: it is possible to have different relative risk budgets for different goals as the asset pricing equations are independent of $\rho(\tau, L)$, while the asset allocations are impacted by this variable.

(d) Multiple Possible Equilibria: While there are no free parameters, it is possible that there could be multiple equilibria for the given conditions (i.e., the equilibrium is not unique). This will be explored in future research.

(e) Entangled Markets or Butterfly Effect: As one can see from the equilibrium equations, the asset returns are inter-linked closely to various parameters, especially correlations and variances of different assets available in the markets. In Muralidhar (2025), we examine what happens to equilibrium asset prices and wages if investors hold "less than optimal" allocations to the risky asset and/or the relative risk-free asset using both Brazilian and US data. These simulations suggest that markets are entangled as one observes in real-life (unlike a CAPM model or Efficient Market Hypothesis, wherein event studies just examine the impact of an "event" on the return of just a single security), and there is a "butterfly effect"; namely, a sub-optimal allocation to even a relative risk-free asset in Brazil could impact the returns of the S&P500 US Equity index.

VII. Conclusion

We believe we have achieved the Cochrane (2022) challenge that "portfolio theory and application should describe a general equilibrium, and how investors are heterogeneous." GBI is slowly becoming the norm for investors, and investors have multiple stochastic goals (and hence offer simple heterogeneity models as opposed to manipulating non-existent utility functions), delegate to agents (that they hope are skillful), budget risk in a precise manner, and seek to maximize goal-relative risk adjusted returns. Countries like Brazil are creating unique instruments to match the cash flow of these goals, and these GADs, has probably opened the door to additional financial innovation that is cash-flow focused. But after showing how this heterogeneity in desired cash flows has led to the creation new securities, we go further in this paper with a more ambitious goal to provide a more robust GE model relative to previous models, that is based on reality.

For example, Arrow-Debreu (AD) create an abstract world with profit maximizing firms, utility maximizing consumers and a fixed number of commodities to establish a competitive equilibrium for the prices of these commodities. While foundational from an academic perspective of asset pricing, AD securities do not and will never exist as there is no natural issuer and “states”/ “commodities” are hard to specify. We present a more realistic normative model, where investors invest to achieve multiple stochastic investment goals like retirement, saving for a child’s education, health etc., but we treat the utility function as “implicit” because we use actual objectives of investors. The creation and issuance of unique and innovative retirement and education bonds by Brazil in 2023, to good and sustained demand, opens a new avenue for finance theory and the ability to test this normative model with live data on these instruments and other generic risky assets.

We show that the existence of just two of these GADs, under certain realistic market conditions and objective functions of investors, along with the absolute risk-free asset, can be used to determine the price of all assets and provide effective asset allocation recommendations for multiple stochastic investment goals in a model we call GRAPM. We demonstrate reasonable prices for given assumptions about assets and, in turn, the wages that must be earned by investors/consumers/workers to hold such portfolios because of the budget constraint to do so. Moreover, we can also specify a generic (implicit) production function, such that the firms generating such desired cash flows, paying appropriate wages to the investors/consumers/workers, could be treated as profit maximizers and their securities priced in the same competitive general equilibrium. We also include a “welfare”-focused government that supplies the absolute risk-free asset (for either long or short portfolios), but where this asset also has a very specific rate of return (and multiple equilibrium equations to ensure “no free parameters”). Hence, this is a much more realistic model than the AD General Equilibrium model (or others) as it is based on positive observations, but with implicit utility (labor allocation) and production functions, and the potential to test how normative models need to be modified with real data to ensure this GE. It also offers new avenues for research as this approach is easily extended by relaxing some of the assumptions made to arrive at this model.

Adam Smith notes, "He is led by an invisible hand to promote an end which was no part his intention."[27] This paper shows how individuals maximizing their goal-specific risk-adjusted return, and likelihood of hiring skillful agents, for a given risk budget and goal, and with unspecified production and labor allocation functions, could lead to a very compact general market equilibrium and pricing model with interesting theoretical and practical implications.

-*-

[27] https://www.panmurehouse.org/adam-smith/smith-quotes-faqs/#:~:text=quotes%20from%20Smith-,'No%20society%20can%20surely%20be%20flourishing%20and%20happy%2C%20of%20which,'

Appendix I – Luck versus skill formula

Outperformance over a benchmark does not tell the investor whether the agent is skillful, even if it is normalized by the relative risk taken to achieve this outperformance (i.e., the Differential Sharpe Ratio). Nor does it provide the investor with a measure of confidence that excess returns were generated by skill-based processes as performance data is noisy. Critical factors involved in answering the luck versus skill question include time, the desired degree of confidence, the investment returns of the portfolio and the benchmark, the standard deviation of the portfolio and the benchmark, and the degree of correlation between the two. The more volatile the portfolio and an agent's excess return series, the greater the noise and, hence, the more the time needed to resolve the issue.

Ambarish and Seigel (1996) frame the question in a truly dynamic world and derive useful equations which can inform investors about the skill of a manager.

. Assume that portfolio (*P*) and the benchmark (L) are the two portfolios under consideration

In finance it is typical to think of risk as variance or the squared standard deviation of returns from the mean return. Therefore, in a relative comparison, one would want to define the variance of the relative return per unit of time as the difference of the portfolio and the benchmark. In other words, one would want the following to hold:

$$\sigma_R^2 = \text{variance per unit of time of } \left(\frac{dP}{P} - \frac{dL}{L}\right) \quad \text{(A.1)}$$

follow the generalized Weiner process so that

$$\frac{dP}{P} = \mu_p dt + \sigma_P dz \quad \text{(A.2)}$$

$$\frac{dL}{L} = \mu_L dt + \sigma_L dz \quad \text{(A.3)}$$

Where (μ_P,σ_P) and (μ_L,σ_L) are the instantaneous mean and volatility parameters of the portfolio and benchmark, respectively. Define $\rho_{P,L}$ as the correlation between *P* and *I* and define the ratio of the portfolio and benchmark as R(t) such that

$$R(t) = \frac{P(t)}{L(t)} \quad \text{(A.4)}$$

The dynamics of R(t) can be extracted using Ito's Lemma such that

$$\frac{dR}{R} = \left(\mu_p - \mu_L + \sigma_L^2 - \sigma_P \sigma_L \rho_{P,L}\right)dt + \sigma_P dz_p - \sigma_L dz_L \quad \text{(A.5)}$$

As noted in the original article, "the variance of the stochastic terms in (A.5) are identical to those in equation (A.1), completing the intuitive appeal of using R(t) as measure of relative performance."

Since we are interested in the time development of R(t), define dz such that

$$\sigma_R dz = \sigma_P dz_p - \sigma_L dz_L \quad \text{(A.6)}$$

Squaring both sides and reducing since $dz^2 = dt$ and $dz_P * dz_L = \rho_{P,L} dt$

$$\sigma_R^2 = \sigma_P^2 + \sigma_L^2 - 2 * \sigma_P * \sigma_L * \rho_{P,L} \quad \text{(A.7)}$$

Equation (A.7) is very interesting as it is exactly the square of the tracking error in equation (7), and the square of the denominator in the Differential Sharpe Ratio. This validates the use of this formulation for the relative risk measure that Sharpe (1994) used.

One can go a step further and derive a simple geometric Brownian motion from (A.5) for R such that

$$R(t) = R(0) * e^{\sigma_R \epsilon \sqrt{t}} * e^{\left[\left\{\left(\mu_P - \frac{\sigma_P^2}{2}\right) - \left(\mu_L - \frac{\sigma_L^2}{2}\right)\right\}t\right]} \quad \text{(A.8)}$$

Where ∈ is the standard normal variable. The first exponential factor in equation (A.8) is the noise component and the second exponent is the true skill in added value. Therefore, for skill to dominate noise, a minimum number of data points are needed or alternatively,

$$T > \frac{K^2 * (\sigma_P^2 - \sigma_L^2 - 2 * \sigma_P * \sigma_L * \rho_{P,L})}{\left\{\left(\mu_P - \frac{\sigma_P^2}{2}\right) - \left(\mu_L - \frac{\sigma_L^2}{2}\right)\right\}^2} \quad \text{(A.9a)}$$

Where *K* is the number of standard deviations for a given confidence interval. For example, when K =1, then one desires an 84% confidence in the skill of the manager.

Through cross multiplication, and taking the square root of both sides, results

$$\sqrt{T} * \left[\frac{\left\{ \left(\mu_P - \frac{\sigma_P^2}{2} \right) - \left(\mu_L - \frac{\sigma_L^2}{2} \right) \right\}}{\sqrt{\left(\sigma_P^2 - \sigma_L^2 - 2 * \sigma_P * \sigma_L * \rho_{P,L} \right)}} \right] > \mathrm{K} \qquad \text{(A.9b)}$$

For each agent, one can input the values on the left-hand side of equation (A.9b) and deduce the confidence in skill. Since (A.9b) isolates the differences in volatility (between the agent's portfolio and benchmark) in the numerator, and normalized for TE in the denominator, it is easy to see why this approach rates agents identical to M-cube. M-cube normalizes for differences in volatility and correlation (i.e., together TE).

Appendix II – Derivation of M-cube Risk Adjusted Performance

Principals will

max E[r(A) – r(L)] or equation (1) (A.10)

subject to

$E[r(A)] = a^{P}_{\tau/L} \times E[r(P)] + l^{L}_{\tau/L} \times E[r(L)] + (1 - a^{P}_{\tau/L} - l^{L}_{\tau/L}) \times r(F)$ (A.11)

$\sigma(A) = \sigma(L)$ (A.12)

and TE(A, L) = TE(Target, L) = τ (A.13)

where $a^{P}_{\tau/L}$ is the allocation to the risky (agent) portfolio, P, given the goal of L and a target relative risk of T; $l^{L}_{\tau/L}$ is the allocation to the goal-hedging/goal-replicating asset, L (and measures what is *de-facto* invested in the low-cost passive benchmark or "beta" in industry parlance). The super-script denotes the asset being allocated to; the subscript indicates the target relative risk (τ) and the goal-replicating asset (L). The balance of the assets is invested in the traditional absolute risk-free asset, F (and measures leverage).

Now

$TE(A, L) = \sqrt{[\sigma^2(A) - 2 * \rho(A, L) * \sigma(A) * \sigma(L) + \sigma^2(L)]}$ (A.14)

Where ρ(*) is the correlation parameter. From the constraint on tracking error (equation 6 or A.13), a unique target correlation between portfolio A and liability L, $\rho(\tau, L)$, is identified. Alternatively, instead of specifying a TE(Target, L), a principal could also specify $\rho(\tau, L)$.

Substituting (A.14) in (A.13) and squaring both sides, and substituting for $\sigma(A) = \sigma(L)$ obtains

$\rho(\tau, L) = 1 - \frac{TE(Target, L)^2}{2 \times \sigma^2(L)} = 1 - \frac{\tau^2}{2 \times \sigma^2(L)}$ (A.15)

The optimal allocations to "a" and "l" ensures that $\rho(A,L)$ equals $\rho(\tau, L)$ and $\sigma(A) = \sigma(L)$.

Further, calculating $\sigma^2(L) = \sigma^2(A)$ using equations (A.11) and (A.12) gives

$$\sigma^2(L)=(a^P_{\tau/L})^2\times\sigma^2(P) + (\, l^L_{\tau/L})^2\times\sigma^2(L) + 2\times(a^P_{\tau/L}) \times (\, l^L_{\tau/L}) \times\sigma(P) \times\sigma(L) \times\rho(P,L) \qquad (A.16)$$

But, the covariance of A and L, can be expressed below using the definition of covariance and equation (A.11) as

$$\sigma(A) \times \sigma(L) \times \rho(A,L) = a^P_{\tau/L} \times \sigma(P) \text{ x } \sigma(L) \times \rho(P,L) + \, l^L_{\tau/L} \times \sigma^2(L) \qquad (A.17)$$

Using equation (A.12), and imposing $\rho(A,L) = \rho(\tau,L)$ implied from equation (A.13) as an optimal condition, equation (A.17) can be re-written as

$$\rho(T,L) \times \sigma^2(L) = a^P_{\tau/L} \times \sigma(P) \text{ x } \sigma(L) \times \rho(P,L) + \, l^L_{\tau/L} \times \sigma^2(L) \qquad (A.18)$$

Dividing the RHS and LHS by $\sigma^2(L)$ and re-arranging terms,

$$l^L_{\tau/L} = \rho(\tau, L) - a^P_{v/L} \times \frac{\sigma(\mathrm{P})}{\sigma(\mathrm{L})} \times \rho(P,L) \qquad (A.19)$$

Substituting (A.19) in (A.18) and solving gives

$$a^P_{\tau/L} = \frac{\sigma(L)}{\sigma(P)}\left[\sqrt{\frac{[1-\rho(\tau,L)^2]}{[1-\rho(P,L)^2]}}\right] = \frac{\sigma(L)}{\sigma(P)}\left[\frac{\varphi(\tau,L)}{\varphi(P,L)}\right], \text{ where } \varphi(I,J) = \sqrt{[1-\rho(I,J)^2]} \qquad (A.20)$$

And substituting (A.19) in (A.20) gives

$$l^L_{\tau/L} = \rho(\tau,L) - \rho(P,L) \times \left[\sqrt{\frac{[1-\rho(\tau,L)^2]}{[1-\rho(P,L)^2]}}\right] = \rho(\tau,L) - \rho(P,L) \times \left[\frac{\varphi(\tau,L)}{\varphi(P,L)}\right] \qquad (A.21)$$

Appendix III – Path to Deriving GRAPM

Define $E[r(A)_{x,y|}\ \tau]$ to be the expected risk-adjusted return of portfolio *A* with *x* as the risky asset and *y* as the goal, with τ as the target relative risk. Recall equations (A.20) and (A.21) establish optimal allocations to risky, goal-replicating and absolute risk-free (*F*) assets for investors maximizing goal-relative expected risk-adjusted returns, given target absolute and relative risk levels. At this stage, this is an optimal, but a non-equilibrium result.

1. First consider goal *G1* in isolation. Use equation (4) to establish the expected risk-adjusted return of a portfolio for the *G1* goal with *I* as risky asset, and goal-replicating asset *G1* and define it as $E[r(A)_{I,G1|}\ \tau]$. This is shown in equation (A.22) which is a repeat of the M-cube formulation.

$$E[r(A)_{I,G1|\tau}] = \left\{\frac{\sigma(G1)}{\sigma(I)}\left[\frac{\varphi(\tau,G1)}{\varphi(I,G1)}\right]\right\} \times E[r(I)] + \left\{\rho(\tau,G1) - \rho(I,G1) \times \left[\frac{\varphi(\tau,G1)}{\varphi(I,G1)}\right]\right\} \times E[r(G1)] +$$

$$\left(1 - \frac{\sigma(G1)}{\sigma(I)}\left[\frac{\varphi(\tau,G1)}{\varphi(I,G1)}\right] - \left\{\rho(\tau,G1) - \rho(I,G1) \times \left[\frac{\varphi(\tau,G1)}{\varphi(I,G1)}\right]\right\}\right) \times r(F) \qquad (A.22)$$

2. Similarly, establish the expected risk-adjusted return of a portfolio for the *G1* goal with *G2* as risky asset and define it as $E[r(A)_{G2,G1|}\ \tau]$. This is shown in equation (A.23).

$$E[r(A)_{G2,G1|}\tau] = \left\{\frac{\sigma(G1)}{\sigma(G2)}\left[\frac{\varphi(\tau,G1)}{\varphi(G2,G1)}\right]\right\} \times E[r(G2)] + \left\{\rho(\tau,G1) - \rho(G2,G1) \times \left[\frac{\varphi(\tau,G1)}{\varphi(G2,G1)}\right]\right\} \times$$
$$E[r(G1)] + \left(1 - \frac{\sigma(G1)}{\sigma(G2)}\left[\frac{\varphi(\tau,G1)}{\varphi(G2,G1)}\right] - \left\{\rho(\tau,G1) - \rho(G2,G1) \times \left[\frac{\varphi(\tau,G1)}{\varphi(G2,G1)}\right]\right\}\right) \times r(F) \qquad (A.23)$$

3. This is the most important step and statement in the paper. If a risk-adjusted portfolio has the same volatility as the goal (*G1*) and the same target correlation ($\rho(\tau, G1)$), then it cannot have two expected values based on which asset was used to create it (unless there is an arbitrage opportunity or a market imperfection). In other words, the equilibrium condition requires that $E[r(A)_{I,G1|}\ \tau] = E[r(A)_{G2,G1|}\ \tau]$.[28] This equality

[28] See Appendix V for 3 approaches to justify this assumption.

between equations (A.22) and (A.23) allows us to derive $E[r(I)_{G1}]$ as a function of *G1*, *G2* and *F* as in equation (A.27). For convenience, we will dispense with τ in the subscript as the terms below are independent of τ.

If $a'_{G1} = \frac{\sigma(I)}{\sigma(G2)}\left[\frac{\varphi(I,G1)}{\varphi(G2,G1)}\right]$ (A.24)

And $l'_{G1} = \frac{\sigma(I)}{\sigma(G1)} \times \rho(I,G1) - \frac{\sigma(I)}{\sigma(G1)} \times \rho(G2,G1) \times \left[\frac{\varphi(I,G1)}{\varphi(G2,G1)}\right]$ (A.25)

Define $E[r(I)_{G1}]$ as the expected return of asset *I* in *G1*. Then, setting equation (A.22) = (A.23) gives

$$E[r(I)_{G1}] = a'_{G1} \times E[r(G2)] + l'_{G1} \times E[r(G1)] + (1 - a'_{G1} - l'_{G1}) \times r(F) \quad \text{(A.26)}$$

Alternatively, equation (A.26) can be expressed in the following manner by rearranging terms.

$$E[r(I)_{G1} - r(F)] = a'_{G1} \times E[r(G2) - r(F)] + l'_{G1} \times E[r(G1) - r(F)] \quad \text{(A.27)}$$

It may appear that equation (A.27) offers the asset pricing equation desired, but this is only a partial result because it is based entirely on just one set of investors – those with *G1* as the goal or the myopic perspective. So far, there is no limit on the parameters. It is the heterogeneity of investors/goals that gives the full equilibrium and ensures no "free parameters".

4. Similarly, for the *G2* goal, follow steps 1-3 to derive $E[r(A)_{I,G2}]$ and $E[r(A)_{G1,G2}]$. Once again, as in Step 4, set $E[r(A)_{I,G2}] = E[r(A)_{G1,G2}]$ to derive $E[r(I)_{G2}]$ as function of *G1*, *G2* and *F* as in equation (A.31).

   If $a''_{G2} = \frac{\sigma(I)}{\sigma(G1)}\left[\frac{\varphi(I,G2)}{\varphi(G1,G2)}\right]$ (A.28)

   And $l''_{G2} = \frac{\sigma(I)}{\sigma(G2)} \times \rho(I,G2) - \frac{\sigma(I)}{\sigma(G2)} \times \rho(G1,G2) \times \left[\frac{\varphi(I,G2)}{\varphi(G1,G2)}\right]$ (A.29)

   Then

$$E[r(I)_{G2}] = a''_{G2} \times E[r(G1)] + l''_{G2} \times E[r(G2)] + (1 - a''_{G2} - l''_{G2}) \times r(F) \qquad \text{(A.30)}$$

Or alternatively, rearranging terms in equation (A.30),

$$E[r(I)_{G2} - r(F)] = a''_{G2} \times E[r(G1) - r(F)] + l''_{G2} \times E[r(G2) - r(F)] \qquad \text{(A.31)}$$

5. Now assume the representative investors across the two goals can trade freely. We assume that asset *I* cannot have two different returns for investors with goal *G1* and goal *G2*, unless there is a market inefficiency. In other words, an equilibrium between the two sets of representative investors requires that $E[r(I)_{G1}] = E[r(I)_{G2}]$ (or equation (A.27) = equation (A.31)).

   Recall that there are three classes of assets and GRAPM provides specific expected returns for each asset. This condition gives two key equilibrium conditions: equation (A.34) for r(*F*) and equation (A.35) for the relationship between goal-replicating assets, *G1* and *G2*.

   If we define

$$X_{I,G1,G2} = \varphi(I,G1) - \rho(I,G2) \times \varphi(I,G1) + \rho(G1,G2) \times \varphi(I,G2) \qquad \text{(A.32)}$$

$$Y_{I,G1,G2} = \varphi(I,G2) - \rho(I,G1) \times \varphi(I,G2) + \rho(G1,G2) \times \varphi(I,G1) \qquad \text{(A.33)}$$

   Then

$$r(F)_{G1,G2} = \frac{\left\{E[r(G1) - \frac{\sigma(G1)}{\sigma(G2)} \times (\frac{X_{I,G1,G2}}{Y_{I,G1,G2}}) \times E[r(G2)]\right\}}{\left[1 - \frac{\sigma(G1)}{\sigma(G2)} \times (\frac{X_{I,G1,G2}}{Y_{I,G1,G2}})\right]} \qquad \text{(A.34)}$$

   Alternatively, equation (27) can be restated as

$$E[r(G1) - r(F)] = \frac{\sigma(G1)}{\sigma(G2)} \times (\frac{X_{I,G1,G2}}{Y_{I,G1,G2}}) \times E[r(G2) - r(F)] \qquad \text{(A.35)}$$

   Using equations (A.28), (A.29) and (A.35) in equation (A.31) yields the key asset pricing equation (A.36) for risky asset *I*, using *G1* as a risky asset and *G2* as the goal.

$$\mathrm{E}[\mathrm{r}(I) - \mathrm{r}(F)] = \mathrm{Z}_{I,G1/G2} \times \mathrm{E}[\mathrm{r}(G1) - \mathrm{r}(F)] \tag{A.36}$$

Where Zeta is a form of a "covariance term" as beta is in CAPM, defined as

$$\mathrm{Z}_{I,G1/G2} = \frac{\sigma(I)}{\sigma(G1)} \times \left\{ \left[\frac{\varphi(I,G2)}{\varphi(G1,G2)}\right] + \left(\frac{Y_{I,G1,G2}}{X_{I,G1,G2}}\right) \times \rho(I,G2) - \left(\frac{Y_{I,G1,G2}}{X_{I,G1,G2}}\right) \times \rho(G1,G2) \times \left[\frac{\varphi(I,G2)}{\varphi(G1,G2)}\right] \right\} \tag{A.37}$$

Notice, that Zeta is a function of two volatilities, $\sigma(I)$ and $\sigma(G1)$, (as in CAPM) but three correlations: $\rho(G1, G2)$, $\rho(I, G2)$ and $\rho(I, G1)$. One other key point to note, equation (A.36) cannot be used for assets that are goal-replicating; instead, equation (A.35) must be used for goal-replicating assets. Further, since X and Y are functions of $\varphi(I, J)$, Zeta is not defined when $\rho(G1, G2)$, $\rho(I, G2)$ or $\rho(I, G1) = 1$, unless the other correlations are = 0.

Interestingly, Zeta can be restated in terms of "*a*" and "*l*" using equations (13) and (14) as follows and will be interpreted later:

$$\mathrm{Z}_{I,G1/G2} = \frac{a^{G1}_{\tau/G2}}{a^{I}_{\tau/G2}} + \frac{\sigma(I)}{\sigma(G1)} \times \left(\frac{Y_{I,G1,G2}}{X_{I,G1,G2}}\right) \times l^{G2}_{I/G2} \tag{A.38}$$

6. Similarly, express E[r(*I*)] as a function of E[r(*G2*)], r(*F*), volatilities and correlations (assuming *G1* is the goal) as in equation (A.39).

$$\mathrm{E}[\mathrm{r}(I) - \mathrm{r}(F)] = \mathrm{Z}_{I,G2/G1} \times \mathrm{E}[\mathrm{r}(G2) - \mathrm{r}(F)] \tag{A.39}$$

In a way, we can see these dual equations in (A.36) and (A.39) for I as a pair-wise or "Janus" equilibrium.[29]

7. Interestingly, given equations (A.39), (A.35) and (A.36), there is a unique relationship between $\mathrm{Z}_{I,G1/G2}$ and $\mathrm{Z}_{I,G2/G1}$; namely, that

[29] In this example, the Janus duality is demonstrated by the fact that *I* can be priced in this pair-wise equilibrium with *G1* as risky (*G2* as goal) or vice versa. This duality is unchanged even when we add more goal-replicating assets, as we only use two at a time to create the asset pricing equation for *I*.

$$Z_{I,G2/G1} = \frac{\sigma(G1)}{\sigma(G2)} \times \left(\frac{X_{I,G1,G2}}{Y_{I,G1,G2}}\right) \times Z_{I,G1/G2} \tag{A.40}$$

Appendix IV – The Three Goal-Replicating GRAPM

With three goals; namely, *G1, G2* and *G3*, there are now six potential equations for the expected returns of asset *I*. In equilibrium, all must provide the same result because all assets are linked by the "pair-wise" equilibrium.

$$E[r(I) - r(F)] = Z_{I,G3/G1} \times E[r(G3) - r(F)] \quad (A.41)$$

$$E[r(I) - r(F)] = Z_{I,G3/G2} \times E[r(G3) - r(F)] \quad (A.42)$$

$$E[r(I) - r(F)] = Z_{I,G1/G3} \times E[r(G1) - r(F)] \quad (A.43)$$

$$E[r(I) - r(F)] = Z_{I,G2/G3} \times E[r(G2) - r(F)] \quad (A.44)$$

Interestingly, given equations (A.39) and (A.44) and the fact that Zeta is applied to the same risk premium, $E[r(G2) - r(F)]$, and continuing in a similar fashion for all other goal-replicating assets (in this case $E[r(G1) - r(F)]$ and $E[r(G3) - r(F)]$), then is must be the case that

$$Z_{I,G2/G1} = Z_{I,G2/G3} \quad (A.45)$$

From equations, (A.41) and (A.42), we can derive equation (A.46).

$$Z_{I,G3/G1} = Z_{I,G3/G2} \quad (A.46)$$

And from equations (A.36) and (A.43), we can derive equation (A.47).

$$Z_{I,G1/G2} = Z_{I,G1/G3} \quad (A.47)$$

Notice that equations (A.45), (A.46) and (A.47) are a bit different from equation (A.40) because in these equations the second goal-replicating asset is identical on both sides of the equation (e.g., *G2* in equation A.45), but not so in equation (A.40). Using the approach to derive equation (A.40) could provide additional relationships among the Zetas given the three-goal case.

This "Janus" duality of Zeta is interesting and in contrast to the single "beta" one derives from CAPM. Further, given the equilibrium condition in equation (A.35) between two goal replicating assets, we can additionally show that,

$$E[r(G1) - r(F)] = \frac{\sigma(G1)}{\sigma(G3)} \times (\frac{X_{I,G1,G3}}{Y_{I,G1,G3}}) \times E[r(G3) - r(F)] \tag{A.48}$$

$$E[r(G2) - r(F)] = \frac{\sigma(G2)}{\sigma(G3)} \times (\frac{X_{I,G2,G3}}{Y_{I,G2,G3}}) \times E[r(G3) - r(F)] \tag{A.49}$$

Using equation (A.49) in (A.35), we can eliminate E[r($G2$) – r(F)] and derive equation (A.50).

$$E[r(G1) - r(F)] = \frac{\sigma(G1)}{\sigma(G3)} \times (\frac{X_{I,G1,G2}}{Y_{I,G1,G2}}) \times (\frac{X_{I,G2,G3}}{Y_{I,G2,G3}}) \times E[r(G3) - r(F)] \tag{A.50}$$

Setting equation (A.50) equal to equation (A.43), gives an additional equilibrium equation; namely, that

$$\left(\frac{X_{I,G1,G3}}{Y_{I,G1,G3}}\right) = \left(\frac{X_{I,G1,G2}}{Y_{I,G1,G2}}\right) \times \left(\frac{X_{I,G2,G3}}{Y_{I,G2,G3}}\right) \tag{A.51}$$

From these equations we can also show that

$$\left(\frac{X_{I,G2,G3}}{Y_{I,G2,G3}}\right) = \left(\frac{Y_{I,G3,G2}}{X_{I,G3,G2}}\right) \tag{A.52}$$

And similarly for the other combination of $G1$, $G2$ and $G3$.

Adding more goal replicating assets gives not only more equilibrium equations for the generic risky asset I, but also additional equalities for Zeta and relationships between various Xs and Ys. What this implies is that the equilibrium has to be very compact across all variables; namely, that each asset in the market probably can only take limited expected returns, volatilities and correlations to ensure this full market equilibrium. Hence our claim that there are probably few "free" parameters in this model. As shown mathematically above, we can also conclude that,

vi. For each risky asset I, if there are N goal-replicating assets, then there will be N x (N-1) equations for I.

vii. There will be {N x (N-1)}/2 equilibrium equations for goal-replicating assets.

viii. For each I, there will be N equalities for Z.

ix. For each I, there will be N-1 equations that express the equilibrium between the X and Y variables as in (25) and (26).

x. There are also N equations for F.

Appendix V – Proof of Key Assumption in the Paper

In this Appendix we provide three proofs for why the key assumption in the paper is acceptable; namely, that

$$E[r(A)_{I,G1|}\,\tau] = E[r(A)_{G2,G1|}\,\tau] \qquad \text{(A.53).}$$

This Appendix addresses this assumption through three methods: (a) an Intuitive Argument; (b) A Specific Case; and (c) the General Result.

a. *An Intuitive Argument*

If $E[r(A)_{I,G1|}\,\tau] < E[r(A)_{G2,G1|}\,\tau]$, then one could argue that no principal would hire the agent that has created a portfolio with asset *I*, and hence with zero demand, $E[r(I)] = 0$.

Also, the equality can hold with *I* and *G2* being completely different assets. Note that $a^{I}_{\tau/G1} \neq a^{G2}_{\tau/G1}$ and similarly, for the allocation to *G1* in both risk-adjusted portfolios, because *I* and *G2* are assumed to be distinct assets.

b. *A Specific Case*

One may try to argue that $E[r(A)_{I,G1}] = E[r(A)_{G2,G1}]$ is not guaranteed when the two portfolios have the same absolute and relative risk, even with an identical correlation to the goal. One possibility suggested by a reader is that if r(*I*), r(*G2*), r(*G1*) have the same variance, but are independent, then this may not be true. However, if we plug in $\sigma(I) = \sigma(G2) = \sigma(G1)$ and if $\rho(G2, G1) = \rho(I, G1) = \rho(G2, I) = 0$ into our (A.53), then it results in $E[r(I)] = E[r(G2)]$, which violates our assumption that the risky assets are unique. As a result, we must rule out this special case.[30]

[30] Thanks to Prof. Kazuhiko Ohashi for this insight.

c. The General Result

The third option starts with the expanded form of both sides of equation (A.53). From (A.22) and (A.23), assume that (A.53) is not true. Then,

$$\{\frac{\sigma(G1)}{\sigma(I)}\left[\frac{\varphi(\tau,G1)}{\varphi(I,G1)}\right]\} \times \mathrm{E}[r(I)] + \{\rho(\tau,G1) - \rho(I,G1) \times \left[\frac{\varphi(\tau,G1)}{\varphi(I,G1)}\right]\} \times \mathrm{E}[r(G1)] + (1 - \frac{\sigma(G1)}{\sigma(I)}\left[\frac{\varphi(\tau,G1)}{\varphi(I,G1)}\right] - \{\rho(T,G1) - \rho(I,G1) \times \left[\frac{\varphi(\tau,G1)}{\varphi(I,G1)}\right]\}) \times r(F) \neq$$

$$\{\frac{\sigma(G1)}{\sigma(G2)}\left[\frac{\varphi(\tau,G1)}{\varphi(G2,G1)}\right]\} \times \mathrm{E}[r(G2)] + \{\rho(\tau,G1) - \rho(G2,G1) \times \left[\frac{\varphi(\tau,G1)}{\varphi(G2,G1)}\right]\} \times \mathrm{E}[r(G1)] + (1 - \frac{\sigma(G1)}{\sigma(G2)}\left[\frac{\varphi(\tau,G1)}{\varphi(G2,G1)}\right] - \{\rho(\tau,G1) - \rho(G2,G1) \times \left[\frac{\varphi(\tau,G1)}{\varphi(G2,G1)}\right]\}) \times r(F) \qquad (A.54)$$

Eliminating common terms from both the LHS and RHS.

$$\{\frac{\sigma(G1)}{\sigma(I)}\left[\frac{1}{\varphi(I,G1)}\right]\} \times \mathrm{E}[r(I)] + \{-\rho(I,G1) \times \left[\frac{1}{\varphi(I,G1)}\right]\} \times \mathrm{E}[r(G1)] + \{-\frac{\sigma(G1)}{\sigma(I)}\left[\frac{1}{\varphi(I,G1)}\right] \times \mathrm{r}(F)\} - \{-\rho(I,G1) \times \left[\frac{1}{\varphi(I,G1)}\right]\} \times r(F) \neq$$

$$\{\frac{\sigma(G1)}{\sigma(G2)}\left[\frac{1}{\varphi(G2,G1)}\right]\} \times \mathrm{E}[r(G2)] + \{-\rho(G2,G1) \times \left[\frac{1}{\varphi(G2,G1)}\right]\} \times \mathrm{E}[r(G1)] + \{-\frac{\sigma(G1)}{\sigma(G2)}\left[\frac{\varphi(\tau,G1)}{\varphi(G2,G1)}\right]\} \times r(F) - \{-\rho(G2,G1) \times \left[\frac{\varphi(\tau,G1)}{\varphi(G2,G1)}\right]\} \times r(F) \qquad (A.55)$$

$$\{\frac{\sigma(G1)}{\sigma(I)}\left[\frac{1}{\varphi(I,G1)}\right]\} \times \mathrm{E}[r(I) - \mathrm{r}(F)] - \rho(I,G1) \times \left[\frac{1}{\varphi(I,G1)}\right]\} \times \mathrm{E}[r(G1) - \mathrm{r(F)}] \neq$$

$$\{-\rho(G2,G1) \times \left[\frac{1}{\varphi(G2,G1)}\right]\} \times \mathrm{E}[r(G1) - \mathrm{r(F)}] \qquad (A.56)$$

Which boils down to equation (A.57), which we could add as an additional assumption to the model to ensure that the model is robust.

$$\sigma(G1)\left[\frac{\mathrm{E}[r(I) - \mathrm{r}(F)]}{\sigma(I)\varphi(I,G1)} - \frac{\mathrm{E}[r(G2) - \mathrm{r(F)}]}{\sigma(G2)\varphi(G2,G1)}\right] \neq \left[\frac{\rho(G2,G1)}{\varphi(G2,G1)} - \frac{\rho(I,G1)}{\varphi(I,G1)}\right] \times \mathrm{E}[r(G1) - \mathrm{r(F)}] \qquad (A.57)$$

Using the assumptions in Appendix V.b., this results in $\mathrm{E}[\mathrm{r}(I)] \neq \mathrm{E}[\mathrm{r}(G2)]$.

Appendix VI – The Various Equilibrium Conditions for an Implicit Production Function and Labor Allocation Process

We first begin with the equilibrium condition/budget constraint of the two individuals, S_{G1} and S_{G2}

$\omega_{G1} = \omega_{G1}^{G1} + \omega_{G1}^{G2} + \omega_{G1}^{I}$ = Value of Portfolio *G1* (A.58)

$\omega_{G2} = \omega_{G2}^{G1} + \omega_{G2}^{G2} + \omega_{G2}^{I}$ = Value of Portfolio *G2* (A.59)

Similarly, each of the three entities will need to pay wages to these workers, but also under the zero economic profit assumption, and the existence of an unspecified production function (e.g., for *G1*, $\emptyset_{\boldsymbol{G1}}(G1, \text{ʖ}, \omega_{G1}^{G1}, \omega_{G2}^{G1}, \text{Þ}(\boldsymbol{G1}))$, where ʖ is some capital used to produce *G1*, and by applying labor of S_{G1} and S_{G2} and paying them wages, and so on for *G2* and *I*), we can derive equations A.60 – A.62

$\omega^{G1} = \omega_{G1}^{G1} + \omega_{G2}^{G1} = \text{Þ}(G1)$*Amount of Shares Sold (*G1*) (A.60)

$\omega^{G2} = \omega_{G1}^{G2} + \omega_{G2}^{G2} = \text{Þ}(G2)$*Amount of Shares Sold (*G2*) (A.61)

$\omega^{I} = \omega_{G1}^{I} + \omega_{G2}^{I} = \text{Þ}(I)$*Amount of Shares Sold (*I*) (A.62)

where "*Amount of Shares Sold" is the column provided in Table 4d, which combines the asset allocations of both investors to the three key assets.

Combining these 5 equations and solving, we must have the GE condition that

Value of Portfolio *G1* + Value of Portfolio *G2* = $\text{Þ}(G1)$*Amount of Shares Sold (*G1*) + $\text{Þ}(G2)$*Amount of Shares Sold (*G2*) + $\text{Þ}(I)$*Amount of Shares Sold (*I*) (A.63)

What we are obviously missing in addition to specific production functions for each of the entities, is a clear allocation rule for S_{G1} to allocate their labor to entities E_{G1}, E_{G2} and E_{I}, and similarly for S_{G1}.

Appendix VII. Testing the GRAPM/GE Model with 3 GADs (Using Real Data)

We add RendA+ 2035 to the mix as G3, and demonstrate the results first without any adjustment (which does not converge). Then, as we did in the 2 GAD case, we modify one of the correlation assumptions and appear to converge closer to a solution, but the message from the 3 or N GAD model is that various inputs cannot be as arbitrary as they are in a CAPM world. Hence, more robust solving techniques are needed than just Microsoft EXCEL Solver.

We benchmark these results to the results in Table 4.a as shown in Table A.VII.a below

	ASSET PRICING							
	Expected Return	Volatility	Return/Risk	Correlation				Price
Asset				F	S&P500 TR	RENDA 2060	EDUCA 2030	
	2	3	4	5	6	7	8	
F	2.02%	0		1	0	0	0	0.9802
S&P500 TR	8.29%	13.5%	0.61	0	1	-0.1417	-0.142	0.9234
RENDA 2060	4.72%	36.0%	0.13	0		1	0.873	0.9550
EDUCA 2030	2.55%	7.1%	0.36	0			1	0.9751

Target Tracking Error		3%
ρ(τ,G1)		0.997
ρ(τ,G2)		0.911

Table A.VII.a. The 2 GADs Case with 3% Target Tracking Error and Solving for r(*F*), r(*RENDA 2060*), r(*EDUCA 2030*), and S&P500 in BRL terms.

We compare this initially to the results using live data on volatilities and correlations in Table A.VII.b, which as noted earlier is not a total equilibrium as not all Zeta conditions are met.

	ASSET PRICING								Price
	Expected Return	Volatility	Return/Risk	Correlation					
	2	3	4	5	6	7	8	9	
Asset				F	S&P500 TR	RENDA 2060	EDUCA 2030	RENDA 2035	
F	2.88%	0		1	0	0	0	0	0.9720
S&P500 TR	8.74%	13.5%	0.65	0	1	-0.162	-0.142	-0.148	0.9196
RENDA 2060	5.32%	36.0%	0.15	0		1	0.873	0.934	0.9495
EDUCA 2030	3.37%	7.1%	0.47	0			1	0.936	0.9674
RENDA 2035	3.90%	14.9%	0.26					1	0.9625

Table A.VII.b. The 3 GADs Case with 3% Target Tracking Error and Solving for r(*F*), r(*RENDA 2060*), r(*EDUCA 2030*), r(*RENDA 2035*) and S&P500 in BRL terms.

Also, adding RendA 2035 once again provides a realistic result as it has a higher duration than Educa2030 and hence earns a higher return (in an upward-sloping yield curve environment). Even though this is not an equilibrium result, we can see that the values for all three classes of assets are different. So, this indicates that testing this model might be as hard as testing CAPM as we will never know how many assets are goal-replicating (i.e., relative safe asset) and which are truly risky (e.g., *I*). For completeness, we show a slightly modified version of this base case with live data, to show how the results from the base case might change as we change the input variables and this is shown in Table A.VII.c.

	ASSET PRICING								
	Expected Return	Volatility	Return/Risk	Correlation					Price
	2	3	4	5	6	7	8	9	
Asset				F	S&P500 TR	RENDA 2060	EDUCA 2030	RENDA 2035	
F	2.89%	0		1	0	0	0	0	0.9719
S&P500 TR	8.74%	13.5%	0.65	0	1	-0.070	-0.142	-0.148	0.9196
RENDA 2060	5.34%	36.0%	0.15	0		1	0.873	0.934	0.9493
EDUCA 2030	3.36%	7.1%	0.47	0			1	0.936	0.9675
RENDA 2035	3.86%	14.9%	0.26					1	0.9628

Table A.VII.c. The 3 GADs Case with 3% Target Tracking Error and Solving for r(*F*), r(*RENDA 2060*), r(*EDUCA 2030*), r(*RENDA 2035*) and S&P500 in BRL terms, with One Modified Correlation.

The good news in comparing the last two tables is that the expected returns/prices are not markedly different (a few basis points of difference), and hence while one may want to seek the ultimate GE solutions with all Zeta equilibrium conditions being met, the marginal benefit to expected return estimates is trivial. Muralidhar (2024b) shows how in some cases, modifying the volatility assumption ensures convergence, which we have not attempted to do here, given the small benefit to estimates.

Acknowledgements

The author would like to thank the late Prof. Franco Modigliani and Lester Seigel for providing the inspiration for the three key pillars of this paper. Also, thanks to Jean-Luc Vila for helpful comments and the nudge to pursue this paper. We derive a lot of inspiration from the works (and friendship) of Paul Pfleiderer and John Cochrane.

References

Ai, H., M. M Croce, K. Li. 2013. Toward a Quantitative General Equilibrium Asset Pricing Model with Intangible Capital. *The Review of Financial Studies*, Volume 26, Issue 2, February 2013, Pages 491–530, https://doi.org/10.1093/rfs/hhs121

Allen, F. 2001. Do Financial Institutions Matter? *Journal of Finance* 56, no. 4 (August): 1, 165–1,175.

Ambarish, R., and L. Seigel. 1996. Time is the Essence. *Risk* 9, no. 8 (August): 41–42.

Arrow, K. J., 1953. The role of securities in the optimal allocation of risk– bearing. As translated and reprinted in 1964, *Review of Financial Studies*, (31), 91–96.

Arrow, K. J. and G. Debreu. 1954. Existence of an Equilibrium for a Competitive Economy. *Econometrica*, Vol. 22, No. 3. (Jul., 1954), pp. 265-290.

Black, F. and R. Litterman. (1992). Global Portfolio Optimization, *Financial Analysts Journal*, September/October, 48(5): 28-43.

Black, F. and R. Litterman. 1991. Asset Allocation: Combining Investor Views with Market Equilibrium, *Journal of Fixed Income*, September: 7-18.

Bodie, Z., R.C. Merton and W.F. Samuelson. 1992. Labor supply flexibility and portfolio choice in a life cycle model, *Journal of Economic Dynamics and Control,* Volume 16, Issues 3–4, July–October 1992, Pages 427-449

Brennan, M. 1993. Agency and Asset Pricing. *Working paper, University of California Los Angeles Anderson Graduate School of Management*.

CASSEL, G, 1924, *The Theory of Social Economy,* New York: Harcourt, Brace and Company, 1924.

Chan, Y. L. and L. Kogan. (2002). "Catching Up with the Joneses: Heterogeneous Preferences and the Dynamics of Asset Prices," *Journal of Political Economy* 110(6), 1255–1285.

Cochrane, J. 2005. *Asset Pricing (revised edition). Princeton University Press. Princeton, NJ. https://bsutrisno.files.wordpress.com/2017/02/cochrane_2005_asset-pricing.pdf*

Cochrane, J. H. 2022. Portfolios for Long-Term Investors. *Review of Finance*, Volume 26, Issue 1, February 2022, Pages 1–42, https://doi.org/10.1093/rof/rfab038

Coqueret, G. L. Martellini, and V. Milhas. 2017. Equity Portfolios with Improved Liability Hedging Benefits. *Journal of Portfolio Management*, Volume 43, Issue #2, pp 37-49.

Cornell, B. *2009.* Luck, Skill, and Investment Performance, *The Journal of Portfolio Management*, Winter 2009, 35 (2), 131 - 134

Cornell, B., and *R.* Roll. 2005. *A* Delegated Agent Asset-Pricing Model. *Financial Analysts Journal,* 61, no. 1 (January/February): 57–69.

Cox, J.C., J. Ingersoll. S. Ross. 1985. An Intertemporal General Equilibrium Model Of Asset Prices, Econometrica 53(2):363-84; DOI: 10.2307/1911241

Das, S., H. Markowitz, J. Scheid, and M. Statman. 2010. Portfolio Optimization with Mental Accounts. *Journal of Financial and Quantitative Analysis*, Vol. 45, No. 2 (2010), pp. 311-334

Debreu, G., 1959. Theory of Value: An Axiomatic Analysis of Economic Equilibrium. Yale University Press, New Haven.

Gomes, J. F. and L. Schmid. 2017. Equilibrium Asset Pricing with Leverage and Default. *Unpublished Working Paper, https://faculty.wharton.upenn.edu/wp-content/uploads/2016/03/JF-Submission.pdf.*

Graham, J.R. and C.R. Harvey. 1994. Market Timing Ability and Volatility Implied in Investment Newsletters' Asset Allocation Recommendations. *NBER Working Paper #4890*, October 1994

Graham, J.R. and C.R. Harvey. 1997. Grading the Performance of Market Timing Newsletters. *Financial Analysts Journal,* 53(6):54-66.

Lucas, R. 1978. Asset prices in an exchange economy. *Econometrica*, 46:1429–1445, December 1978.

M. Haugh, and A. Lo. 2001. Asset allocation and derivatives. *Quantitative Finance*. Volume 1, #1, January 2001.

Markowitz, H. 1952. Portfolio Selection. *Journal of Finance.* 7, no. 1 (March): 77–91.

Merton, R. 1973. Intertemporal Capital Asset Pricing Model. *Econometrica* 41, no. 5 (September): 867–887.

Merton, R. C. (1983). "On Consumption Indexed Public Plans," In *Financial Aspects of the United States Pension System*, Z. Bodie and J. Shoven (eds.). National Bureau of Economic Research. Cambridge, MA. (Chapter 18 in *Continuous-Time Finance*. Reprinted in *The Foundations of Pension Finance*, Volume I, Z. Bodie and E. P. Davis, eds., Edward Elger, 2000.).

Merton, Robert C. 1993. "Optimal Investment Strategies for University Endowment Funds." In *Studies of Supply and Demand in Higher Education,* edited by C. Clotfelter and M. Rothschild. Chicago: University of Chicago Press, 1993. (Chapter 21 in Continuous-Time Finance.)

Merton, R., 2014. The Crisis in Retirement Planning. *Harvard Business Review*, July – August 2014. http://hbr.org/2014/07/the-crisis-in-retirement-planning/ar/1

Merton. R. C. 2007. "The Future of Retirement Planning." In T*he Future of Life-Cycle Saving & Investing*, Z. Bodie, D. McLeavey, and L.B. Siegel (eds.). Research Foundation of the CFA Institute, 2007, Charlottesville, VA.

Merton, R. C., and A. Muralidhar. 2020. SeLFIES: A New Pension Bond and Currency for Retirement. *Journal of Financial Transformation,* April 2020.

Modigliani, Franco, and A. Ando. 1963. The Lifecycle Hypothesis of Savings: Aggregated Implications and Tests. *American Economic Review*, 53:55–84.

Modigliani, Franco and Richard Brumberg. 1980. Utility Analysis and the Aggregate Consumption Function: An Attempt at Integration. Unpublished manuscript 1952, published in Abel, A (ed.) The Collected Papers of Franco Modigliani, vol. 12. Cambridge, MA: MIT Press.

Modigliani, Franco, and Merton Miller. 1958. The Cost of Capital, Corporate Finance and The Theory of Investment. *American Economic Review*, 48:261–297.

Modigliani, F., and L. Modigliani. 1997. Risk-Adjusted Performance. *Journal of Portfolio Management,* 23, no. 2 (Winter): 45–54.

Muralidhar, A. 2000. Risk-Adjusted Performance – The Correlation Correction. *Financial Analysts Journal.* 56(5), pp 63-71.

Muralidhar, A. 2001. Optimal Risk-Adjusted Portfolios with Multiple Managers. *The Journal of Portfolio Management*. Spring 2001, Vol. 27, No. 3: pp. 97-104 https://doi.org/10.3905/jpm.2001.319805

Muralidhar, Arun, 2015. The Sharpe Ratio Revisited: What It Really Tells Us (March 18, 2015). Journal of Performance Measurement, Vol 19, No. 3, 2015, Available at SSRN: https://ssrn.com/abstract=2692859

Muralidhar, A. 2016. GBI = Gimme Better Instruments. An Innovation to Simplify Complex Investment Approaches. Investments & Wealth Monitor 31, no. 3: 54–57.

Muralidhar A. 2019. The F-Utility of Wealth: It's All Relative. *Journal Of Investment Management*, Vol. 17, No. 2, (2019), pp. 1–11

Muralidhar, A. 2021. Asset Pricing, Asset Allocation and Risk-Adjusted Performance with Multiple Goals and Agency: The Goals and Risk-based Asset Pricing Model. *Journal Of Investment Management*, Vol. 19, No. 2, (2021), pp. 1–28

Muralidhar, A. 2023. Realativity in Finance: Goals and Risk-based Asset Pricing for Investors with Multiple Goals and Agents. *Journal Of Investment Management*, Vol. 21, No. 3, (2023), pp. 1–29

Muralidhar, Arun. 2024a. Goals-based Arrow-Debreu Securities (December 29, 2024). Available at SSRN: https://ssrn.com/abstract=4579066 or http://dx.doi.org/10.2139/ssrn.4579066

Muralidhar, A. 2024b. Investment Theory and Practice – It's All 'Realative': How Multiple Stochastic Goals and Delegation Changes the Entire Investment Paradigm, August 2024, Publisher: *Georgetown University Open Access* – *https://repository.digital.georgetown.edu/handle/10822/1088947?q=muralidhar*

Muralidhar, Arun. 2025a. The Entangled Markets Hypothesis or The Butterfly Effect in a Normative, General Equilibrium for Asset Prices and Wages (April 21, 2025). Available at SSRN: https://ssrn.com/abstract=5224847 or http://dx.doi.org/10.2139/ssrn.5224847

Muralidhar, Arun. 2025b. A Preferred Habitat, General Equilibrium Model for Asset Prices, Wages and Asset Allocation (and Term Structure) (June 18, 2025). Available at SSRN: https://ssrn.com/abstract=5311230 or http://dx.doi.org/10.2139/ssrn.5311230

Muralidhar, Arun. 2025c. A Realistic, Heterogeneous (Principal-) Agent Asset Pricing Model: With Asset Allocation and Risk-Adjusted Performance (August 23, 2025). Available at SSRN: https://ssrn.com/abstract=5412218 or http://dx.doi.org/10.2139/ssrn.5412218

Muralidhar, A., K. Ohashi, and S. Shin. 2014a. The Relative Asset Pricing Model: Implications for Asset Allocation, Rebalancing, and Asset Pricing. *Journal of Financial Perspectives* (https://www.gfsi.ey.com/the-journal-of-financial-perspectives.php) March 2014.

Muralidhar, A., K. Ohashi, and S. Shin. 2014b. The Relative Asset Pricing Model: Toward a Unified Theory of Asset Pricing, *Journal of Investment Consulting*, Vol. 15, No. 1, 51-66, 2014

Muralidhar, A, and Ohashi, Kazuhiko and Shin, Sung Hwan. 2016. The Most Basic Missing Instrument in Financial Markets: The Case for Bonds for Financial Security (December 1, 2016). Journal of Investment Consulting, Vol. 16, No. 2, p. 34-47, 2016, Available at SSRN: https://ssrn.com/abstract=2880002

Muralidhar, A. and L. Seigel. 2024. Interpretation of Implicit Time in the Sharpe Ratio. *Unpublished Working Paper. Researchgate.net. https://www.researchgate.net/publication/385384912_Interpretation_of_Implicit_Time_in_the_Sharpe_Ratio*

Muralidhar, A., and S. Shin. 2013. The Relative Asset Pricing Model - Incorporating Liabilities and Delegation to Chief Investment Officers: Version 0.1, *Journal of Investment Consulting*, Volume 14, Number 1, 2013, pp 25-40

Muralidhar, A. and Khin Mala U. (1997). Establishing a Peer Comparator Universe for an Institutional Investor, *Journal of Pension Plan Investing*, Spring: 1(4).

Muralidhar, A., and J. W. van Stuijvenberg. 2005. "Devising an Investable Liability Index." *Investments and Pensions—Europe* (October): 46.

Perold, A. 2004. The Capital Asset Pricing Model. *Journal of Economic Perspectives,* Volume 18, Number 3—Summer 2004—Pages 3–24

Pfleiderer, P. 2020. Chameleons: The Misuse of Theoretical Models in Finance and Economics, *Economica*, 87, #345, January 2020, Pages 81-107

Roll, Richard. 1996. "U.S. Treasury inflation-protected bonds: the design of a new security." *Journal of Fixed Income* 6, 9–28.

Sharpe, W. 1964. *Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk. Journal of Finance 19, no. 3 (September): 425–442.*

Sharpe, W. 1994. The Sharpe Ratio. *Journal of Portfolio Management* 21, no. 1 (fall): 49–58.

Sharpe, W. and L. Tint. 1990. Liabilities—A New Approach. *Journal of Portfolio Management* 16, no. 2 (winter): 5–10.

Shiller, Robert. 2005. "In Favor of Growth-Linked Bonds." *The Indian Express.* March 10.

Shiller, Robert (1993). "Macro Markets: Creating Institutions for Managing Society's Largest Economic Risks". New York: Oxford University Press.

Tobin, James. 1958. Liquidity Preference as Behavior Towards Risk. *Review of Economic Studies* 25, no. 2 (February): 65–86.

van Binsbergen, Jules H. and Brandt, Michael W. and Koijen, Ralph S. J., Optimal Decentralized Investment Management (October 19, 2006). EFA 2006 Zurich Meetings, AFA 2007 Chicago Meetings Paper, Available at SSRN: https://ssrn.com/abstract=887399 or http://dx.doi.org/10.2139/ssrn.887399

WALD, A.1951. "Uber einige Gleichungssysteme der mathematischen Gkonomie," *ZeitschrijtfGr Nationalokonomie,* Vol. 7 (1936), pp. 637-670, translated as "On Some

Systems of Equations of Mathematical Economics," *ECONOMETRICA*, Vo1., 19, October 1951, pp. 368403.